INCH-POUND

MIL-DTL-32403
11 December 2012

DETAIL SPECIFICATION
ITEM SPECIFICATION
FOR THE

FIRE EXTINGUISHER, EMPLOYING HFC-227ea AGENT ENHANCED WITH SPECIAL SODIUM BICARBONATE POWDER (HFC-227ea/SBC_S), PORTABLE, WITH BRACKET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Identification. This document establishes the requirements and verification methods for an HFC-227ea/SBC_S Hand-held Fire Extinguisher (HHFE) with Bracket. For simplicity, the term 'HHFE system' will be used to describe a fully-charged HFC-227ea/SBC_S HHFE assembly and its mounting bracket. The term 'HHFE' will most often describe a charged or uncharged HHFE assembly without the bracket. The HHFE can consist of a Steel Pressure Cylinder or an Aluminum Pressure Cylinder (paragraph 3.3.1).

1.2 System Overview. The HFC-227ea/SBC_S HHFE system is comprised of the following items: a non-refillable, one-time usage HHFE (see paragraph 3.4.2); the fire extinguishing agent (paragraph 3.2.2); the pressurization gas (paragraph 3.2.4); and the point-of-use mounting bracket (paragraph 3.4.4). The fire extinguishing agent HFC-227ea/SBC_S is comprised of 1, 1, 1, 2, 3, 3, 3-Heptafluoropropane agent (generically named HFC-227ea) having specialized sodium bicarbonate (SBC_S) powder in suspension. The purpose of this HHFE system is to provide Army Aviation weapon systems with an acceptable HHFE that does not employ an extinguishing agent classified as an ozone-depleting substance (ODS).

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army Research, Development and Engineering Center, ATTN: RDMR-SET, Redstone Arsenal, AL 35898-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications standard and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

RR-C-901	Cylinders, Compressed Gas: Seamless Shatterproof, High Pressure DOT 3AA Steel, and 3AL Aluminum
BB-E-2879	Extinguisher, Fire, Carbon Dioxide (CO2), Portable, with Mounting Bracket, for Aviation Use
TT-C-490	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

FEDERAL STANDARDS

FED-STD-595/11105	Red, Gloss
FED-STD-595/36638	Gray, Gloss
FED-STD-595/37031	Black Miscellaneous Lusterless

COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-59503	Technical Nitrogen

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-32412	HFC-227ea Fire Extinguishing Agent Enhanced with Special Sodium Bicarbonate Powder (HFC-227ea/SBC_S)
MIL-PRF-22750	Coating, Epoxy, High-Solids
MIL-DTL-53072	Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspections
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

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DEPARTMENT OF DEFENSE STANDARDS

MILITARY

MIL-STD-810	Environmental Engineering Considerations & Laboratory Tests
MIL-STD-1472	Human Engineering

(Copies of these documents are available on-line at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS
Title 49 - Transportation

Part 178.35	General requirements for specification cylinders
Part 178.46	Specification 3AL seamless aluminum cylinders
Part 178.50	Specification 4B welded or brazed steel cylinders

(Copies of these documents are available on-line at http://www.gpo.gov/fdsys/search/home.action)

U.S. ARMY AVIATION AND MISSILE COMMAND

HHFE Assembly	Technical Data Package (TDP), Assembly drawing number 1042950

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), International

ASTM A751	Steel Products, Practices, and Terminology for Chemical Analysis Of
ASTM A959	Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels
ASTM D3335	Standard Test Method for Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy

(Copies of these documents are available on-line at http://www.astm.org)

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UNDERWRITERS' LABORATORY

UL Standard 711	Fire Extinguisher, Rating and Fire Testing of
UL Standard 2129	Standard for Halocarbon Clean Agent Fire Extinguishers
UL Standard 8	Water Based Agent Fire Extinguishers
UL Standard 154	Carbon Dioxide Fire Extinguishers

(Copies of these documents are available from Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096 or http://www.ul.com)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQZ1.4	Sampling Procedures and Tables for Inspection by Attributers

(Copies of ANSI/ASQZ1.4 are available online at www.asq.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First Article. When specified in the contract or order (see paragraph 6.5), sample(s) shall be subjected to first article inspection in accordance with (IAW) paragraph 4.2.

3.2 Materials and components. The materials and components shall conform to applicable specifications, standards, and patterns required herein. Materials not specified shall be selected by the contractor but shall be subject to all provisions of this specification.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Fire extinguishing agent (HFC-227ea/SBC_S). The fire extinguishing agent shall be HFC-227ea/SBC_S Class I or II conforming to specification MIL-DTL-32412 and Appendix A of this document. Verification procedures are outlined in paragraph 4.5.1.1.

3.2.3 Agent exposure. All HHFE internal components to be in contact with the extinguishing agent while it is in storage or in its standby configuration shall consist of materials that are compatible with the agent and/or its constituents for at least 5 years (service life). Compatibility with the agent means that the material shall not degrade such that it loses its functionality nor cause the agent to lose its effectiveness due to contamination from corrosion. Verification procedures are detailed in paragraph 4.5.1.2.

3.2.4 Pressurization gas. The pressurization gas shall be nitrogen which conforms to the purity and moisture requirements of Commercial Item Description (CID) A-A-59503 Type I, Grade A, class 1. Verification procedures are detailed in paragraph 4.5.1.3.

3.2.5 Coatings and surface finishes. Coatings containing Cadmium, Lead, or Hexavalent Chromium shall not be used. Coatings and pretreatments shall be IAW TT-C-490. The painted surfaces of the HHFE shall be red in color, conforming to FED-STD-595, color number 11105. Ferrous bracket surfaces shall be painted interior aircraft black conforming to FED-STD-595, color number 37031. The remaining non-ferrous stainless steel bracket component materials do not require painting. Verification procedures are detailed in paragraph 4.5.1.4. Coatings shall conform to MIL-DTL-53072 (see paragraph 6.4).

3.2.6 Discharge handle. The components of the discharge handle shall be constructed of 16 gauge 304 stainless steel conforming to the composition properties of ASTM A959 Table 1. The nominal thickness of the handle shape material that is unformed (flat) shall be 1.59 mm ± 0.152 (0.0625 inches ± 0.006). Verification procedures are detailed in paragraph 4.5.1.5.

3.2.7 Pull pin/Safety locking device. The safety pull pin/locking device for the discharge handle shall be made of 316 Stainless Steel conforming to ASTM A959 paragraph “Harmonized Standard Grade Stainless Steel Compositions”, Table 1 – Chemical Composition Limits. Verification procedures are detailed in paragraph 4.5.1.6.

3.2.8 Horn. The discharge horn shall be fabricated of electrically nonconductive material. Verification procedures are detailed in paragraph 4.5.1.7.

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3.3 Construction.

3.3.1 Cylinder. The HHFE shall be built using a new hydrostatically tested cylinder with less than six months from date of its manufacture when delivered to the procuring organization. The HHFE's cylinder shall be constructed IAW one of the following:

3.3.1.1 Steel (Type I) cylinder. A steel (Type I) cylinder shall conform to Title 49 Department of Transportation (DOT) Code of Federal Regulations (CFR) Parts 178.50 (DOT-4B) and 178.35 (general requirements). The cylinder's service pressure shall be 500 pounds per square inch (psi) at ambient temperature and be marked accordingly on the cylinder. Verification procedures are detailed in paragraph 4.5.2.1.1.

3.3.1.2 Aluminum alloy (Type II) cylinder. An aluminum alloy (Type II) cylinder shall conform to CFR 49 Parts 178.46 (DOT-3AL) and 178.35 (general requirements) and shall also meet the shatterproof (NONSHAT) requirements for aluminum cylinders as stated in RR-C-901. The cylinder's service pressure shall be no less than 600 psi at ambient temperature and be marked accordingly on the cylinder along with the required "NONSHAT" marking. Verification procedures are detailed in paragraph 4.5.2.1.2.

3.4 Design.

3.4.1 HHFE system. The HHFE System's overall height and width shall conform to FIGURE 1 according to the cylinder material used (see paragraph 3.3.1). With the HHFE clamped in the bracket, its safety pin installed, and its horn positioned alongside the cylinder, the HHFE system using a Type I cylinder shall have an overall height of 33.68 ± 0.51 centimeters (cm) (13.26 ± 0.2 inches) as measured from the bottom of the bracket to the tallest point on the upper handle. From the top view, the overall width shall be 16.99 ± 0.64 cm (6.69 ± 0.25 inches) as measured from the furthest points on the horn and the lower handle. While a Type II cylinder configuration has not been developed to meet the requirements of this specification, it is possible one could be developed in the future. Dimensions for a Type II configuration shall have the same dimensions as the Type I configuration. Verification procedures are detailed in paragraph 4.5.3.1.

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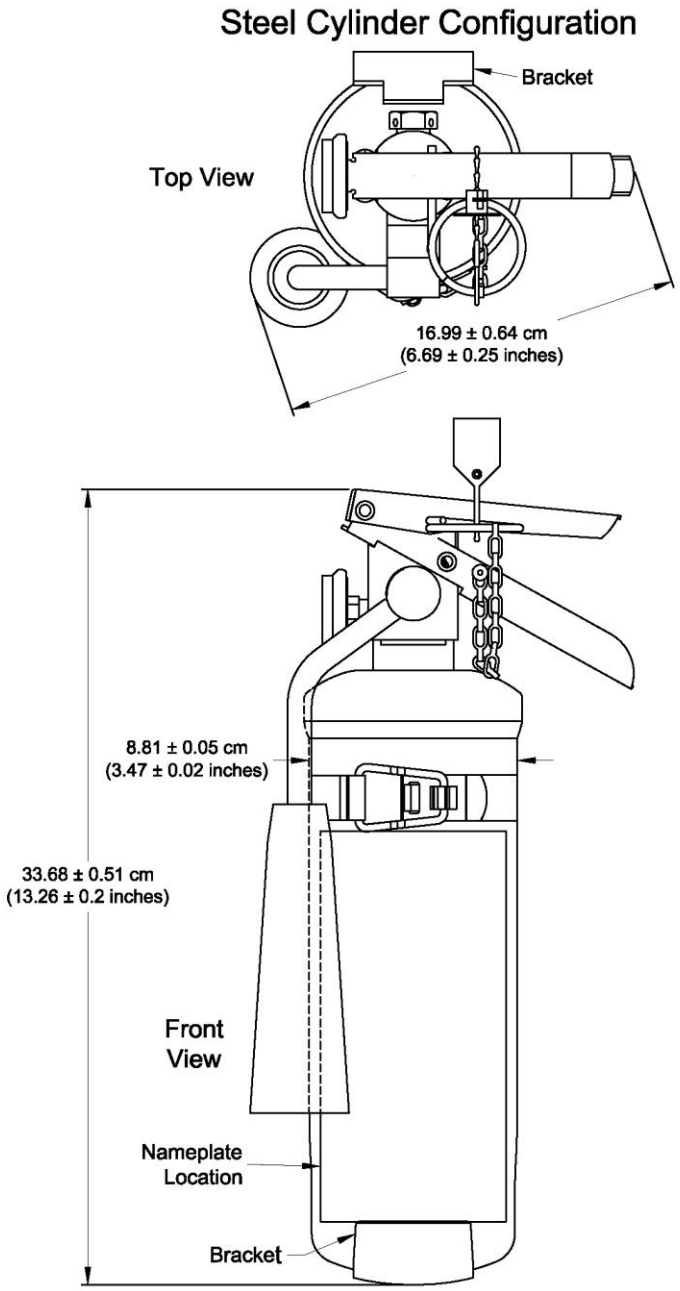


FIGURE 1. HHFE System Steel (Type I) Cylinder Configurations

3.4.2 HHFE. The Type I HHFE's overall height shall be 33.3 ± 0.3 cm (13.12 ± 0.1 inches). The HHFE's design shall provide a means for it to be clamped securely into its bracket and also into the following legacy bracket(s) (NSNs 4210-01-509-8185 (Part number 130-010), 4210-01-261-5118 (Part Number FH-900-26) or 4210-01-519-0943 (Part Numbers 1299-4292 and 130-042)) (see FIGURE 2). Note: Refer to paragraphs 3.5.2 and 3.5.3 for all bracket clamping requirements. Verification procedures are detailed in paragraph 4.5.3.2.

3.4.3 Cylinder. The Type I cylinder shall have an external diameter of 8.81 ± 0.05 cm (3.47 ± 0.02 inches) at the bracket latching location, where the measurement does not include the label (see FIGURE 1). The Type I cylinder with siphon tube installed shall have an internal volume of

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1310 +33, -16 cubic cm (CC) (80 +2, -1 cubic inches). Verification procedures are detailed in paragraph 4.5.3.3.

3.4.4 Bracket. The bracket's overall dimensions and mounting hole locations shall conform to FIGURE 2. The bracket's clamp with the spring lock closed shall retain the HHFE without any sliding or rattling relative to the clamp band when subjected to the acceleration forces specified herein (refer to paragraph 3.5.2 for bracket clamping requirements). The bracket shall also provide an interface to allow it to be attached with fasteners to a structure, such as an aircraft fuselage or other similar surface which will support the total weight of the bracket and the fully-charged HHFE. Verification procedures are detailed in paragraph 4.5.3.4.

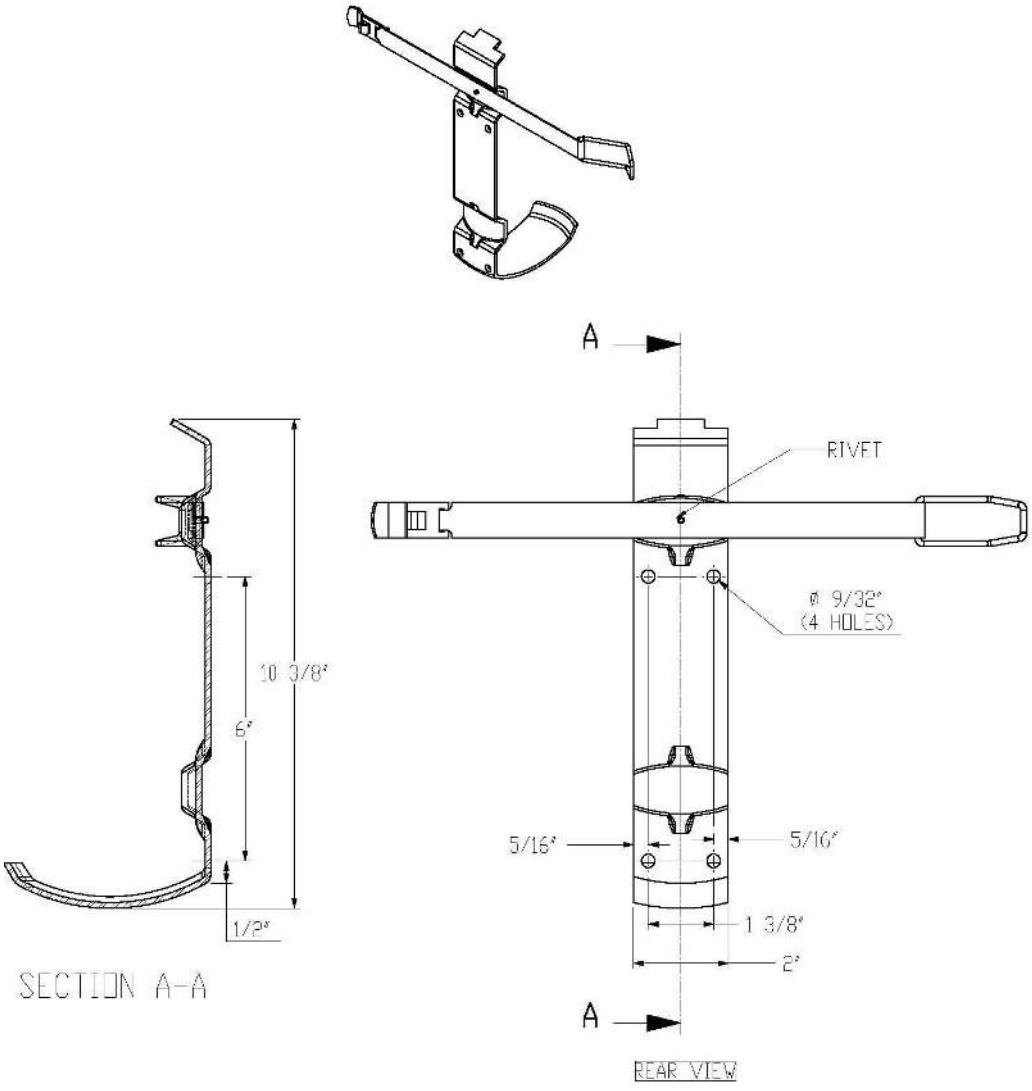


FIGURE 2. Bracket Assembly (NSN: 4210-01-509-8185, 4210-01-261-5118, or 4210-01-519-0943)

3.4.5 HHFE contents and pressurization. If Class I of the extinguishing agent is used (see paragraph 3.2.2), the fully-charged HHFE shall contain 1425 ± 35 grams of the agent at 70°F , and the ullage shall be pressurized to 215 ± 5 psi at 70 ± 2°F with pressurization gas (paragraph 3.2.4). If Class II of the extinguishing agent is used, the fully-charged HHFE shall contain 1416 ±

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35 grams of the agent, and the ullage shall be pressurized to 185 ± 5 psi at 70 ± 2°F with pressurization gas (paragraph 3.2.4). Prior to charging, the HHFE shall be purged with the pressurization gas or a vacuum drawn. Verification procedures are detailed in paragraph 4.5.3.5.

3.4.5.1 SBC_S concentration in HFC-227ea. The percent weight of the Class I SBCs within the charged HHFE shall be 5 ± 0.5%, while the percent weight of the Class II SBCs within the charged HHFE shall be 3.2 ± 0.2%. Verification procedures are detailed in Appendix A.

3.4.6 Discharge indicator gauge. The HHFE shall have an analog pressure gauge with three distinct visual intervals indicated with graduated zones colored as follows: **red** (0 PSI to 100 PSI/-65°F), **green** (100 PSI/-65°F to 700 PSI/160°F), and **red** (700 PSI/160°F to approximately 1000 PSI). Each colored zone transition shall be labeled in text with the corresponding information stated above, but the maximum pressure shall be omitted. Additionally, the HHFE's nominal pressure and temperature of 200 PSI/70°F shall be indicated with a graduated mark and labeled with text. Refer to FIGURE 3 below. Verification procedures are detailed in paragraph 4.5.3.6.

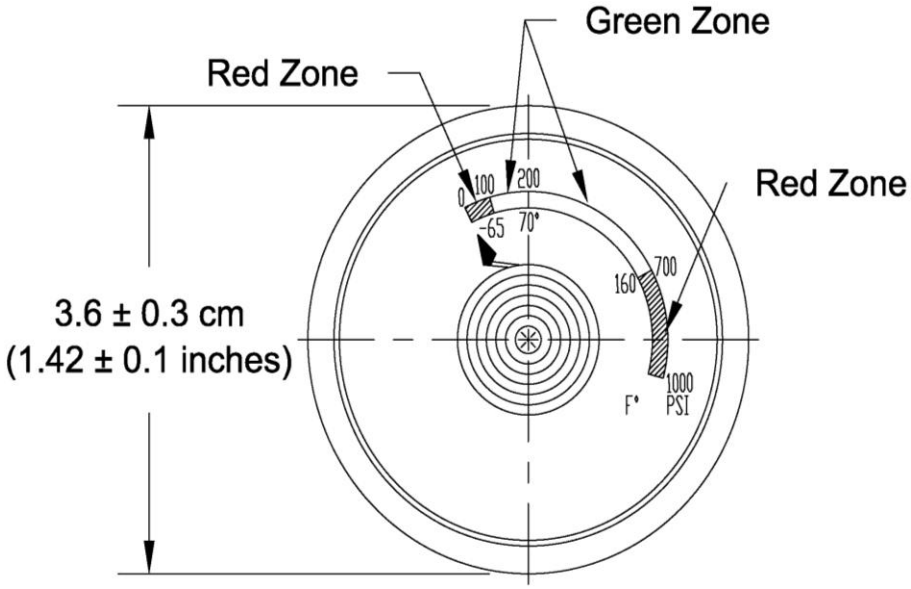


FIGURE 3. HHFE Gauge

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3.4.7 Mounting bracket latch. The bracket holding the extinguisher shall have no more than one latch to release the HHFE from the bracket's clamping mechanism. The HHFE shall provide a means to be removed by hand from any mounted orientation of the bracket when the latch is released. Verification procedures are detailed in paragraph 4.5.3.7.

3.4.8 HHFE removal. The HHFE system shall not cause the HHFE to have removal interference with other items on the aircraft that are mounted in proximity to it. The HHFE's bracket, with the bracket's band unlatched, shall not interfere with the HHFE's extraction from the mounting bracket itself. With the bracket's retention band unlatched, the overall system design shall provide a means for the operator to remove the HHFE by sliding it longitudinally in the direction of its valve not more than 15.2 mm (0.6 inch) away from the bracket base and then extracting it away from the back of the bracket. Verification procedures are detailed in paragraph 4.5.3.8.

3.4.9 Nameplate. The operating, inspection, and maintenance instructions shall be in the form of a pressure-sensitive nameplate attached to the side (painted surface) of the HHFE's cylinder. The nameplate shall have black borders and read in the line sequence shown in FIGURE 4 having the character sizes and colors indicated. Line group 1 through 12 and line group 14 through 29 shall each be printed in black font onto a light gray background matching FED-STD-595 color 36628 (Gray). Lines 13 and 30 shall have light gray characters also matching color 36628 and shall be printed onto the black borders enclosing the two backgrounds. Black font and black borders shall match FED-STD-595 color 37031 (Interior Aircraft Black). The top of the label shall be adhered to the cylinder so that when the HHFE is in the mounted configuration, the bracket strap does not cover any portion of the label. The nameplate shall remain intact, readable, and legible, and shall not exhibit cracking or curling at the edges after being subjected to the exposures described in UL Standard 2129 clauses titled "Nameplate Exposure Tests", "Nameplate Adhesion Test", and "Nameplate Abrasion Test". Verification procedures are detailed in paragraph 4.5.3.9.

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LINE NUMBER	CHARACTER SIZE	COLOR	WEIGHT	TEXT
1	3/16	BLACK	BOLD	TO OPERATE
2	5/32	BLACK		1. PULL RING PIN AND ADJUST HORN
3	5/32	BLACK		2. POINT HORN CLOSE TO BASE OF THE FIRE
4	5/32	BLACK		3. DEPRESS LEVER FOR DISCHARGE AND KEEP
5	5/32	BLACK		BASE OF FLAME COVERED SWEEPING SIDE TO
6	5/32	BLACK		SIDE PUSHING FIRE OFF THE FUEL SOURCE
7	3/16	BLACK	BOLD	CAUTION
8	5/32	BLACK		1. AVOID BREATHING SMOKE
9	5/32	BLACK		2. VENTILATE VEHICLE AFTER DISCHARGE
10	5/32	BLACK		3. AVOID CONTACT WITH SKIN OR CLOTHING
11	5/32	BLACK		4. DO NOT DISCHARGE TOWARD FACE
12	5/32	BLACK		5. CONTENTS ARE UNDER PRESSURE
13	3/16	LIGHT GRAY	BOLD	FIRE EXTINGUISHER, HFC-227ea/SBC, 3 LB
14	3/16	BLACK		FOR MAINTENANCE
15	7/64	BLACK		1. REPLACE WITH A NEW CHARGED EXTINGUISHER, PART
16	7/64	BLACK		NUMBER 1042950 or 1042950-1, IMMEDIATELY AFTER USE OR 5 YEARS
17	7/64	BLACK		AFTER THE DATE OF MANUFACTURE
18	7/64	BLACK		2. REPLACE ANYTIME THE GAUGE READING IS IN THE RED
19	7/64	BLACK		ZONE WHEN AMBIENT TEMPERATURES ARE BETWEEN -65°F
20	7/64	BLACK		AND 160°F
21	7/64	BLACK		3. WEIGH EXTINGUISHER EVERY 12 MONTHS AND REPLACE IF
22	7/64	BLACK		GROSS WEIGHT HAS DECREASED BY THREE OUNCES OR MORE
23	7/64	BLACK		4. INVERT AND VIGOROUSLY SHAKE THE EXTINGUISHER FOR 30
24	7/64	BLACK		SECONDS AFTER COMPLETION OF THE 12 MONTH WEIGHT INSPECTION
25	3/32	BLACK		“NATIONAL STOCK NUMBER (NSN)”
26	3/32	BLACK		“MANUFACTURER'S NAME AND CAGE CODE”
27	3/32	BLACK		“CONTRACT NO.”
28	3/32	BLACK		DATE OF MANUFACTURE "MONTH/YEAR"
29	3/32	BLACK		EXTINGUISHER GROSS WEIGHT "____"LB "____"OZ WITHOUT BRACKET
30	3/16	LIGHT GRAY	BOLD	PROPERTY OF THE U.S. GOVERNMENT

Note: Text within quotation marks is a descriptor of the text or digits to be assigned.

FIGURE 4. Nameplate Instructions

3.4.10 Weight. When charged with agent and pressurization gas IAW paragraph 3.4.5, the HHFE, excluding the bracket, shall weigh no more than 3.6 kilograms (7 pounds, 14 ounces). Verification procedures are detailed in paragraph 4.5.3.10.

3.5 Safe Design.

3.5.1 Safe to handle. The extinguisher assembly and mounting bracket shall be free of sharp edges or sharp surfaces that could possibly injure personnel during maintenance, inspection or use. Verification procedures are detailed in paragraph 4.5.4.1.

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3.5.2 Non-legacy bracket. The mounting bracket shall hold the fire extinguisher against a static load of 350 pounds downward and an 88-pound load upward along the extinguisher axis when mounted vertically. The bracket shall also hold the extinguisher with a 200-pound outward loading applied to the extinguisher at the strap. These loads shall be applied for not less than one minute. Verification procedures are detailed in paragraph 4.5.4.2.

3.5.3 Legacy bracket compatibility. The HHFE shall provide a means for it to be clamped and restrained such that the incumbent legacy Halon 1301 HHFE brackets (NSNs 4210-01-509-8185 (Part number 130-010), 4210-01-261-5118 (Part Number FH-900-26) or 4210-01-519-0943 (Part Numbers 1299-4292 and 130-042)) meet the requirements of paragraph 3.5.2. Verification procedures are detailed in paragraph 4.5.4.3.

3.5.4 Cylinder bracket retention - ballistics. The cylinder of a fully-charged HHFE shall remain intact and within the envelope of its clamped bracket after being shot once with a fully tumbled, 0.30-caliber M2 armor-piercing cartridge traveling at a maximum velocity of 865 m/s (2840 ft/s). Pieces smaller than 2 inches in diameter that originate from the projectile's entry or exit areas shall be considered an exception to the above requirement. The requirements of this paragraph shall apply whether the HHFE is shot in any of the following orientations: (1) with the longitudinal axis of the cylinder normal to the line of fire; (2) with the longitudinal axis of the cylinder 45° from normal toward the gun position and the outlet port (neck) of the cylinder facing away from the weapon; (3) with the longitudinal axis of the cylinder parallel to the line of fire with outlet port (neck) facing away from the gun; and (4) with the longitudinal axis of the cylinder normal to the line of fire with the fully tumbled projectile penetrating through the back of the bracket. Verification procedures are detailed in paragraph 4.5.4.4.

3.5.5 Cylinder fragmentation resistance (non-shatterability).

3.5.5.1 Steel (Type I) cylinder. The resulting dimensional size of the Type I HHFE cylinder constructed IAW paragraph 3.3.1.1 shall not exceed 125 percent of its original size (Reference US Army Drawing No. 12994293 – Fire Extinguisher Assembly / DOT-4B-500 steel cylinder) after the unconstrained, fully-charged HHFE is pierced while at gun room ambient temperature by a tumbling .50-caliber W-2 armor-piercing projectile at a minimum velocity of 2600 feet per second. Verification procedures are detailed in paragraph 4.5.4.5.1.

3.5.5.2 Aluminum alloy (Type II) cylinder. The Type II cylinder shall be manufactured IAW paragraph 3.3.1.2 in order to meet requirements for its non-shatterability and shall be indicated as such with the marking "NONSHAT" IAW paragraph 3.6.2.4 of RR-C-901. Verification procedures are detailed in paragraph 4.5.4.5.2.

3.5.6 Safety pull pin/Locking device. The discharge handle shall provide a means such as a safety pull pin/locking device which shall prevent accidental compression of the handle resulting in an unintentional discharge of the HHFE. The safety pull pin/locking device shall not pose foreign object damage (FOD) threat to the aircraft by providing a means for it to remain connected to the HHFE before and after its extraction from the discharge handle. Verification procedures are detailed in paragraph 4.5.4.6.

3.6 Performance.

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3.6.1 Discharge angle and range. The HHFE shall have a function for discharging 90 percent or more of its contents when the cylinder is oriented in the vertical position (0 degrees), and also at an angle of +45° to -45° from the vertical position. At all orientations, the HHFE shall discharge agent to a range of not less than 2.4 meters ± 0.15 meters (8.0 feet ± 0.5 feet) horizontally from the exit plane of the horn. The HHFE shall achieve this range with its cylinder mounted statically and its swing arm and horn affixed to discharge horizontally. Verification procedures are detailed in paragraph 4.5.5.1.

3.6.2 Discharge time and amount. The elapsed time for discharging 90 percent or greater of the agent contents shall be not less than eight seconds and not more than nine seconds while the HHFE is conditioned to and discharged at 21 ± 1°C (70 ± 2°F). When operated at the vertical position, the agent contents shall be fully discharged when the presence of pressurization gas discharge is observed from the horn. Discharge times shall be measured with the HHFE hardware mounted in a static position. Verification procedures are detailed in paragraph 4.5.5.2.

3.6.3 Class B fire extinguishment. Using fully-charged samples for the two practice trials and each of the following six attempts, the HHFE shall extinguish at least two consecutive 12.5-ft^2 pan fires fueled with JP-8 as described in paragraph a. below. In a like manner, the HHFE shall extinguish two consecutive 7.5-ft^2 pan fires fueled with n-heptane as described in paragraph b below. Verification procedures for both are detailed in paragraph 4.5.5.3.

a. JP-8 fueled pan fire. Using a fully-charged HHFE for two practice trials and each of the following six consecutive attempts, the HHFE shall extinguish at least two consecutive pan fires that have undergone a pre-burn period of 60 ± 5 seconds. The pan shall contain JP-8 fuel with a surface area of 1.16 square meters (12.5 square feet) at the time of ignition. The pan fire fuel shall be JP-8 aviation turbine fuel which conforms to MIL-DTL-83133 specifications.

b. Heptane-fueled pan fire. Using a fully-charged HHFE for two practice trials and each of the following six consecutive attempts, the HHFE shall extinguish at least two consecutive pan fires that have undergone a pre-burn period of 60 ± 5 seconds. The pan area shall contain a Heptane fuel with a surface area of 0.70 square meters (7.5 square feet) at the time of ignition. Each test shall be fueled with commercial-grade heptane having the characteristics described in paragraph 8.2.4 of UL Standard 711.

3.6.4 Discharge pattern. The cross-sectional discharge pattern of the agent from a fully-charged HHFE at 21 ±1°C (70 ± 2°F) shall be a minimum average diameter of 45.7 ± 5.1 cm (18 ± 2 inches) at a distance of 1.8 ± .076 meters (6 ± 0.25 feet) from the horn's exit plane. The HHFE shall achieve the aforementioned pattern with its cylinder mounted statically and vertically (0°) and the horn affixed to discharge horizontally (90°). Verification procedures are detailed in paragraph 4.5.5.4.

3.7 Operation. The HHFE system shall meet the operational and use requirements stated in the below subsections.

3.7.1 Human interface. This requirement shall apply to the HHFE system and tasks required for its operation by a 5th percentile female aircrew member up to a 95th percentile male aircrew

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member. This requirement shall also pertain when the aircrew personnel are properly attired in the most cumbersome Air Warrior configuration which shall include wearing properly fitted heavy flight gloves or arctic mittens. The HHFE system shall provide the means for an air crew member to release a fully-charged HHFE from its bracket with one hand using the intended latch mechanism and then remove the HHFE from within the envelope of the bracket, all within 5 seconds. Secondly, the HHFE system shall provide the means for an aircrew member to remove the safety pull pin from the HHFE handle within 4 seconds and then fully compress the discharge handle for at least 9 consecutive seconds by squeezing the discharge handle with one hand. Verification procedures are detailed in paragraph 4.5.6.1.

3.7.2 Bracket unlatching effort. The required force applied to the bracket's latch to unclamp the strap from the HHFE shall not exceed 65 N (15 lb-f). The bracket's unlatching process shall not require more than two motions. Verification procedures are detailed in paragraph 4.5.6.2.

3.7.3 Discharge effort. IAW the recommendations of paragraph 5.1.4.3.2 of MIL-STD-1472 with respect to operators' hand grip requirements, the force required to fully compress the HHFE's discharge handle during a short duration (less than 3 seconds), shall not exceed 165 N (37 lb-f) at any time. The required force also shall not exceed 100 N (22 lb-f) for a sustained duration of over three seconds. Verification procedures are detailed in paragraph 4.5.6.3.

3.7.4 Safety pull pin & tamper indicator. The tamper indicator/seal shall break when subjected to a force of 65 N (15 lb-f) or less as provided by UL Standard 2129. The safety pull pin used to prevent unintentional discharge of the HHFE shall not require a tensile force greater than 135 N (30 lb-f) to disengage it in the intended manner from the HHFE's discharge handle. Furthermore, the pull pin shall not permanently deform or fracture if a pull force of 135 N (30 lb-f) is applied at a 45-degree angle from its intended longitudinal pull axis. Lastly, the pull pin shall not shear or distort when a force of 225 N (50 lb-f) is exerted upon the discharge handle for 30 seconds. Following application of the force for 30 seconds, the safety pull pin shall disengage IAW this paragraph, and the handle shall compress fully and maintain its agent discharge functionality. Verification procedures are detailed in paragraph 4.5.6.4.

3.8 Environmental conditions. The HHFE system shall meet the environmental requirements stated in the below subsections.

3.8.1 Temperature extremes. The non-operational temperature range of the extinguisher assembly shall be -62°C (-80°F) to +85°C (185°F). The operational range of the extinguisher assembly shall be -54°C (-65°F) to +71°C (160°F) with different levels of performance being allowed as indicated in paragraphs 3.8.1.2 and 3.8.1.4 below. There shall be no observed leakage from the extinguisher after its subjection to the non-operational temperature ranges. Verification procedures are detailed in paragraph 4.5.7.1.

3.8.1.1 Non-operational high temperature. The HHFE shall withstand (see definition in section 6.1) seven consecutive 24-hour cycles consisting of an initial HHFE surface temperature of 33°C (91°F), then conditioning its surface to 85°C (185°F), and finally ramping back down to 33°C (91°F). Following exposure to these non-operational high temperature cycles, the HHFE shall meet the performance requirements of paragraph 3.6.3.a with its surface conditioned to any temperature in the range of standard ambient temperature to +71°C (160°F). Also following exposure to non-operational high temperature cycles, the HHFE shall meet the

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performance requirements for discharge and amount IAW paragraph 3.6.2. Verification procedures are detailed in paragraph 4.5.7.1.1.

3.8.1.2 High temperature operation. The HHFE shall maintain the extinguishing performance requirements of paragraph 3.6.3.a with its cylinder having a surface temperature of 71°C (160°F). The temperature/pressure reading on the pressure indicator gauge shall also read accordingly when the HHFE is conditioned to a calibrated maximum operating temperature. Verification procedures are detailed in paragraph 4.5.7.1.2.

3.8.1.3 Non-operational low temperature. The HHFE shall withstand (defined by section 6.1) storage at a minimum temperature of -62°C (-80°F) for a minimum period of 72 hours and remain operational when its temperature is increased to -54°C (-65°F). There shall be no visibly or audibly detected leakage from the HHFE when the cylinder's surface temperature has been conditioned to -62°C (-80°F). Verification procedures are detailed in paragraph 4.5.7.1.3.

3.8.1.4 Low temperature operation. With its cylinder surface being conditioned to a surface temperature of -54°C (-65°F), the HHFE shall maintain the extinguishing performance requirements of paragraph 3.6.3 with the following exceptions: The HHFE's pan size capability for extinguishing JP-8 and heptane fuels shall be 5 ft^2 and 3 ft^2, respectively. The temperature/pressure reading on the pressure indicator gauge shall also read accordingly when the HHFE is conditioned to a calibrated minimum operating temperature. Verification procedures are detailed in paragraph 4.5.7.1.4.

3.8.1.5 Temperature shock. The HHFE shall maintain the performance requirements for discharge time and amount IAW paragraph 3.6.2 after undergoing multiple cycles of temperature shock. After being gradually conditioned to a temperature of -62°C (-79.6°F), the HHFE shall withstand a minimum of 3 rounds of the following temperature shock cycle: sudden transfer and subsequent conditioning to an environment at +85°C (+185°F), then followed by a sudden transfer and reconditioning to the initial environment of -62°C (-79.6°F). Verification procedures are detailed in paragraph 4.5.7.1.5.

3.8.2 Resistance to high altitude (low pressure) environment. The HHFE's non-operational range of altitude shall be from sea level to 15,240 meters (50,000 feet). As defined by paragraph 6.1, the HHFE shall withstand the atmospheric pressure experienced at the maximum non-operational altitude of 15,240 meters (50,000 feet) for at least one hour, as stated in MIL-STD-810 Method 500.5. Additionally, the HHFE's operational range of altitude shall be from sea level to 6,096 meters (20,000 feet). The HHFE shall operate IAW the performance requirements of paragraph 3.6.2 while exposed to the atmospheric pressure experienced at 6,096 meters (20,000 feet) above sea level. The performance requirement shall be met after at least one hour of exposure to the atmospheric pressure experienced at the maximum operational altitude of 6,096 meters (20,000 feet). Verification procedures are detailed in paragraph 4.5.7.2.

3.8.3 Humidity. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after exposure to a repetitive daily environment having a constant relative humidity of 95 ± 4 percent and a temperature cycle ranging from 30°C (86°F) to 60°C (140°F) and back. Verification procedures are detailed in paragraph 4.5.7.3.

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3.8.4 Fungus. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after being incubated for fungus growth for at least 28 consecutive days at a constant temperature of 30 ± 1°C (86 ± 2°F) and a relative humidity of at least 90 percent but less than 100 percent. Verification procedures are detailed in paragraph 4.5.7.4.

3.8.5 Salt fog. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after exposure to a 5 percent by-weight saltwater spray while the HHFE is conditioned to +35°C (+95°F). The exposure shall be a minimum of two, 24-hour periods of a fog of neutralized saltwater solution at a fallout rate maintained between 1 and 3 milliliters/80 square centimeters/hour. Verification procedures are detailed in paragraph 4.5.7.5.

3.8.6 Sand and dust environment. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after exposure to air having a dust concentration of 10.6 ± 7 grams/m^3 (0.3 ± 0.2 grams/ft^3) and having a velocity of 8.9 ± 1.3 meters/second (1750 ± 250 feet/minute). The environment shall consist of the HHFE being in the direct path of air having a sand concentration of 2.2±0.5 grams/m^3 at a velocity in the range of 18-29 m/s (40-65 mph). The exposure time shall be for a period of at least 90 minutes while the assembly is maintained at the high operating temperature stated in paragraph 3.8.1. All blowing dust and blowing sand parameters shall be IAW MIL-STD-810, Method 510.5, Procedure I-Blowing Dust and Procedure II-Blowing Sand. Verification procedures are detailed in paragraph 4.5.7.6.

3.8.7 Acidic atmosphere. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after exposures to an acidic atmosphere equivalent to what is described herein. The acidic atmosphere shall consist of air atomized with a solution consisting of 11.9 mg sulfuric acid and 8.8 mg nitric acid for every 4 liters of solution in distilled water. The exposures, at a minimum, shall consist of four 2-hour periods of acidic atmosphere exposure with the HHFE system conditioned to 35°C (95°F) followed by 7 days of ambient storage. Detailed Acidic atmosphere criteria and evaluation criteria shall be IAW MIL-STD-810, Method 518.1. Verification procedures are detailed in paragraph 4.5.7.7.

3.8.8 Functional shock. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after experiencing functional shock at the levels usually experienced by flight equipment. The HHFE shall perform as stated above after being subjected to the stated functional shocks in each of three orthogonal axes (longitudinal, transverse, and vertical). The HHFE shall not discharge or show signs of leakage of any quantity of its contents when subjected to the aforementioned functional shocks. Verification procedures are detailed in paragraph 4.5.7.8.

3.8.9 Bench handling shock. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after undergoing the usual level of shock associated with typical work bench handling, such as from periodic inspections and weight checks. The HHFE also shall not discharge or show signs of leakage of any quantity of its contents as a result of bench handling. Verification procedures are detailed in paragraph 4.5.7.9.

3.8.10 Crash hazard shock. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after being subjected to survivable crash levels of 40 g or less while mounted in its mounting bracket. The bracket shall not release from its mount nor fail structurally, and the HHFE shall remain clamped within the bracket. The HHFE also shall not discharge or show signs

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of leakage of any quantity of its contents. Verification procedures are detailed in paragraph 4.5.7.10.

3.8.11 Vibration. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after the HHFE, while clamped in its bracket, has been exposed to a vibration spectrum based on the worst-case platform requirements that are indicated by a modal survey of the HHFE system and the expected platforms. These requirements shall be based on the requirements of MIL-STD-810, Method 514.6. When subjected to these vibrations, the HHFE shall withstand them IAW paragraph 6.1 and shall not discharge any of its contents. The HHFE shall also stay clamped within the HHFE bracket. Verification procedures are detailed in paragraph 4.5.7.11.

3.8.12 Drop shock resistance. The HHFE shall perform IAW paragraph 3.6.2 for discharge time and amount after being dropped six times onto 2-inch plywood backed by concrete from a height of four feet. The six drops shall consist of two drops with the longitudinal axis parallel to the surface, two drops with the longitudinal axis perpendicular to the surface with the extinguisher bottom pointing down, and two drops with the longitudinal axis perpendicular to the surface with the extinguisher bottom pointing up. When subjected to these drops, the HHFE shall withstand them IAW paragraph 6.1 and shall not discharge or leak any of its contents from the nozzle and/or components under pressure. The HHFE shall also not leak as a result of damage to any of the components that are under pressure. Verification procedures are detailed in paragraph 4.5.7.12.

3.8.13 Aging resistance of rubber materials (horn). The horn of the HHFE shall not fracture or crack after undergoing the simulated aging process described herein. The simulated aging process shall consist of the horn being maintained at 71°C (160°F) for 24 hours while enclosed in a pressure bomb that is pressurized to a range of 290 to 310 psi-gauge (psig) with oxygen. Verification procedures are detailed in paragraph 4.5.7.13. Horns fabricated of plastic products shall not be subjected to the test specified in 4.5.7.13.

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4. Verification

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (refer to paragraph 6.1.2 for the definition “inspection”):

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on twenty-six complete, preproduction HHFE systems/assemblies to prove prior to starting production that the manufactured system/assembly complies with the requirements of this specification. The HHFEs shall be produced in accordance with the manufacturer’s normal production processes and tooling. This inspection shall include examination IAW section 4.4 on each of the 26 samples and the testing sequence as shown in Table I.

TABLE I. First article testing requirements and sequence

Samples	Testing requirements and sequence			
1 - 8	4.5.7.5 Salt fog	4.5.4.2 Non-legacy bracket	4.5.6.4.2 Discharge handle loading	4.5.5.3 Class B fire extinguishment (Part a)
9 - 16	4.5.7.11 Vibration	4.5.7.12 Drop test	4.5.4.3 Legacy bracket compatibility	4.5.5.3 Class B fire extinguishment (Part b)
17 - 22	4.5.5.1 Discharge angle & range (2 at each angle), 4.5.5.2 Discharge time & amount, 4.5.3.5 HHFE contents and pressurization	4.5.7.13 Aging resistance of rubber materials (horn)	4.5.3.3 Cylinder	
23 - 26	4.5.1.1 Fire extinguishing agent (conformance to Appendix A and MIL-DTL-32412)			

4.3 Conformance inspection. Conformance inspection shall include the examination of section 4.4 and the inspections specified in section 4.5. Unless otherwise specified in section 4.5, sampling of lots/batches or contract quantities for conformance to inspections shall be IAW ANSI/ASQ Z1.4.

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4.3.1 Standard ambient conditions. Unless otherwise specified in the test procedures, all measurements and tests shall be made at standard ambient conditions. Standard ambient conditions shall be as follows:

Temperature:	21 ± 1.0 °C (70 ± 1.8°F)
Relative Humidity (RH):	20 to 80%
Barometric Pressure:	Site pressure

4.4 Examination. Each HHFE or HHFE system, as indicated by the respective requirement, shall be examined for compliance with the requirements specified in sections 3.2 through 3.4 and paragraphs 3.5.1 and 3.5.6. If specific procedures of verification are not stated in Section 4, the requirements' verifications shall be contractor-proposed and Government approved.

4.5 Methods of inspection

4.5.1 Materials and components.

4.5.1.1 Fire extinguishing agent (HFC-227ea/SBC_S). Fully-charged HHFEs shall be verified for compliance with paragraph 3.2.2 and MIL-DTL-32412. Samples of HHFEs' agent contents shall be obtained IAW Appendix A and analyzed in a lab environment to determine the chemical makeup of the contents IAW MIL-DTL-32412. MIL-DTL-32412 analyses verify that the agent slurry and SBC_S meet parameters for moisture content, caking tendency, lumping tendency, flow, suspendability in HFC-227ea, color, particle aspect ratio (length/width), powder surface area, and all particle size requirements. One procedure in Appendix A of this document provides the steps to verify the weight percentage concentration of either Class I or Class II SBC_S in HHFEs. Results indicative of the agent not being IAW Appendix A and MIL-DTL-32412 for either HFC-227ea/SBC_S Class I or II shall constitute a failure.

4.5.1.2 Agent exposure. Testing of a minimum of 12 HHFEs shall be performed to ensure that all components to be in contact with the MIL-DTL-32412 extinguishing agent, while the HHFE is in storage or in its standby configuration, are compatible with the agent and/or its constituents IAW UL Standard 8 clause titled "Agent Exposure Test" sub-clause titled "Disposable (non-rechargeable) fire extinguishers" with the following additional testing. Prior to visual inspections for corrosion and pitting as required IAW UL Standard 8, 8 of the 12 HHFEs not used for the UL Standard 8 pressure tests shall undergo fire performance testing IAW paragraph 4.5.5.3.a (JP8 Fuel) Class B Fire Extinguishment. Inability to pass fire performance testing in addition to testing and inspections cited in UL Standard 8 sub-clause titled "Disposable (non-rechargeable) fire extinguishers" shall be deemed a failure.

4.5.1.3 Pressurization gas. (See paragraph 6.2)

4.5.1.4 Coatings and surface finishes. Laboratory analysis shall be performed to verify that the HHFE system's coatings and surface finishes contain no Cadmium, Lead, or Hexavalent chromium and that all coatings are IAW TT-C-490. For HHFE systems, a visual inspection of the HHFE's cylinder surface and the painted area of the bracket shall be compared to color chips to verify the colors are IAW FED-STD-595, color numbers 11105 and 37031, respectively. These same HHFE systems shall be analyzed for content of Cadmium, Lead and Hexavalent chrome used in their coating system using the test standards shown in Table II. Colors not matching the

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paint chips specified in this section shall constitute a failure. The presence of Lead, Cadmium or Hexavalent Chrome in the coating systems shall constitute a failure.

TABLE II. Coatings and Surface Treatment Verification Analysis

Paint System Component	Pretreatments	Primers	Topcoats
Lead	Not Applicable	ASTM D 3335	ASTM D 3335
Chrome	See Item 6.4	MIL-DTL-53022	MIL-PRF-22750
Cadmium	Not Applicable	Not Applicable	ASTM D 3335

4.5.1.5 Discharge handle. The components of the discharge handle shall be verified to be constructed of 16 gauge 304 stainless steel meeting the composition properties of ASTM A959 Table 1. Composition properties shall be verified per Paragraph 9.2, ASTM A751. The nominal thickness of the material at an area that is unformed shall be measured. The measured thickness of the unformed material (flat area) shall be 1.59 ± 0.152 mm (0.0625 ± 0.006 inches). Discharge handle materials failing to meet these composition properties and/or the material thickness requirement shall be deemed unacceptable.

4.5.1.6 Pull pin/Safety locking device. The pull pin shall be verified to be constructed of 316 stainless steel meeting the composition properties of ASTM A959 Table 1. Composition properties shall be verified per the commonly accepted techniques for routine chemical analysis of steels (paragraph 9.2) as indicated in the Test Methods' section of ASTM A751. A pull pin failing to meet these composition properties shall be deemed unacceptable.

4.5.1.7 Horn. The horn shall be tested IAW UL Standard 154 Clause titled "Discharge Horn Resistance Test" and shall meet the criteria set forth in this clause. Failure of the horn material to meet the criteria set forth in this standard shall not be acceptable.

4.5.2 Construction. Upon receipt by the procurement organization, the stamping on the HHFE's cylinder shall be visually inspected to verify that the cylinder's hydro-test date is less than six months from its manufacturing date shown on the nameplate.

4.5.2.1 Cylinder. The HHFE's cylinder shall be verified to be IAW one of the construction requirements indicated by the subparagraphs within paragraph 3.3.1. The verification method used shall be one of the two methods below corresponding to the type of cylinder construction chosen.

4.5.2.1.1 Steel (Type I) cylinder. If the HHFE's cylinder is made of steel, its construction, testing and third party verification shall be performed as outlined in Title 49 DOT CFR 49 Parts 178.50 (DOT-4B) and 178.35 (general requirements). HHFE cylinders shall be visually inspected for markings indicating its construction is IAW the above DOT-4B steel cylinder requirements. Service pressure shown in the markings shall indicate 500 psi. If the cylinders do not meet the above requirements, the cylinders shall be deemed unacceptable.

4.5.2.1.2 Aluminum alloy (Type II) cylinder. If the HHFE's cylinder is made of aluminum alloy, its construction, testing, and third party verification shall be performed as outlined in CFR 49 Parts 178.46 (DOT-3AL) and 178.35 (general requirements) and Federal Specification RR-C-

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901. HHFE cylinders shall also be visually inspected for markings indicating its construction and "NONSHAT" properties are IAW the above DOT-3AL aluminum cylinder requirements. Service pressure shown in the markings shall be no less than 600 psi. If the cylinders do not meet the above requirements, the cylinders shall be deemed unacceptable.

4.5.3 Design.

4.5.3.1 HHFE system. The overall height of the HHFE system and its conformance to FIGURE 1 shall be verified for compliance with paragraph 3.4.1 using physical measurements. If an HHFE system having a Type I cylinder, as viewed from the front, does not measure 33.68 ± 0.51 cm (13.26 ± 0.2 inches) from the bottom of the bracket to the tallest point on the upper handle, it shall be deemed unacceptable. Also, from a top view of an HHFE system having a Type I cylinder, if its overall width is not 16.99 ± 0.64 cm (6.69 ± 0.25 inches) as measured from the furthest points on the horn and the lower handle, it shall be deemed unacceptable.

4.5.3.2 HHFE. The HHFE's overall height shall be verified for compliance with paragraph 3.4.2 using physical measurements. If the overall height of a Type I HHFE does not measure 33.3 ± 0.3 cm (13.12 ± 0.1 inches), it shall be deemed unacceptable. Refer to paragraphs 4.5.4.2 and 4.5.4.3 for verification of fitment within the respective brackets.

4.5.3.3 Cylinder. Each HHFE tested shall be emptied and verified to paragraph 3.4.3. The external diameter of each HHFE's cylinder shall be verified for compliance with paragraph 3.4.3 using physical measurement in a location above where the HHFE's label is adhered to. If the external diameter of a Type I HHFE's cylinder does not measure 8.81 ± 0.05 cm (3.47 ± 0.02 inches), it shall be deemed unacceptable. The cylinder's volume with valve assembly with siphon tube installed shall also be verified for compliance with paragraph 3.4.3's appropriate volume requirement by first determining and recording the tare weight of the HHFE. The HHFE shall then be completely filled with deionized water, valve assembly with siphon installed, assembly dried off externally, and its gross weight determined and recorded. The weight change due to the HHFE being filled with water shall be used to calculate the cylinder's volume using the known density of water at ambient conditions. If the volume of a Type I cylinder is not 1310 +33, -16 CC (80 +2, -1 cubic inches), it shall be deemed unacceptable.

4.5.3.4 Bracket. Physical measurements shall be used to verify the bracket(s) to paragraph 3.4.4 and the associated FIGURE 2. If the measurements do not conform to those indicated in FIGURE 2, the bracket shall be deemed unacceptable.

4.5.3.5 HHFE contents and pressurization. HHFEs shall be verified for compliance with paragraph 3.4.5. The temperature of each HHFE shall be adjusted to 70 ± 2°F. The pressure shall be verified by visual inspection of the HHFE's pressure gauge (see paragraph 3.4.6). A pressure reading not in the range of 215 ± 5 psi for Class I agent or 185 ± 5 psi for Class II shall constitute a failure of this verification. Next, the weight of each fully-charged HHFE shall be recorded. The extinguishing agent shall then be fully discharged from the HHFE until the pressurization gas begins to escape and no significant agent discharge is evident. Afterward, the HHFE shall be weighed again. Making the assumption that the weight of any remaining pressurization gas is negligible, if the difference between the two weights recorded before and after discharge is not 1425 ± 35 grams for Class I of the agent or 1416 ± 35 grams for Class II of the agent, a failure of this verification shall be indicated.

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4.5.3.6 Discharge indicator gauge. HHFEs shall be verified for compliance with paragraph 3.4.6. Each HHFE's indicator gauge shall be visually inspected and compared to FIGURE 3 in conjunction with the descriptions of paragraph 3.4.6. A gauge's lack of exhibiting the markings as described in paragraph 3.4.6 shall constitute a failure of this verification. Any additional markings shall constitute failure. The measurement of the overall diameter of the gauge shall be measured. Measurements not within the tolerances of FIGURE 3 shall constitute failure.

4.5.3.7 Mounting bracket latch. Brackets shall be verified for compliance with paragraph 3.4.7. Each HHFE bracket shall be visually inspected and verified to have one latch to release the HHFE from the bracket clamp band. The presence of more than one latch for releasing the HHFE from the bracket shall constitute a failure of this verification.

4.5.3.8 HHFE removal. HHFE systems shall be verified for compliance with the requirements of paragraph 3.4.8. A measuring tape or ruler shall be used to measure the length of the outer tab of the mounting bracket that is located in proximity to the HHFE's bottom shoulder. The tab length shall not be over 13 ± 1 mm (0.5 ± 0.04 inches). A measured vertical lift exceeding 15.2 mm (0.6 inch) during the removal demonstration shall constitute a failure of this verification.

4.5.3.9 Nameplate. The nameplates for HHFEs shall be visually inspected for the proper wording, font size, font color, and background colors IAW paragraph 3.4.9 and FIGURE 4. HHFE nameplates shall be prepared for testing IAW UL Standard 2129 clause titled "Nameplate Exposure Tests". After the nameplates have been subjected to the exposure, adhesion and abrasion tests IAW UL Standard 2129, they shall be visually inspected. If any label does not remain intact, readable, and legible, or if it exhibits cracking or curling at the edges or other failure criteria as described in UL Standard 2129, it shall be considered a failure of this verification.

4.5.3.10 Weight. HHFEs shall be verified for compliance with paragraph 3.4.10. Each HHFE, excluding the bracket, shall be weighed. A weight result exceeding 3.6 kilograms (7 pounds, 14 ounces) shall constitute a failure of this verification.

4.5.4 Safe Design.

4.5.4.1 Safe to handle. The handling safety IAW paragraph 3.5.1 shall be verified by visual inspection. The perceived presence of sharp points, edges, or surfaces that could puncture or cut personnel shall constitute a failure of this verification.

4.5.4.2 Non-legacy bracket. HHFE brackets shall be verified for compliance with paragraph 3.5.2 following the procedures in the below sub-paragraphs.

4.5.4.2.1 Longitudinal loading. The HHFE shall be mounted vertically in each bracket and a downward static load (along the HHFE's longitudinal axis) of 350 pounds applied to the HHFE and an upward 88-pound load along the same axis. The loads shall be applied for one minute. A bracket's inability to retain the HHFE shall constitute a failure.

4.5.4.2.2 Radial loading. A 200-pound outward load (radial to the HHFE) shall be applied to the HHFE at the bracket's strap for one minute. A bracket's inability to retain the HHFE shall constitute a failure.

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4.5.4.3 Legacy bracket compatibility. The HHFE and one of the three available Halon 1301 HHFE legacy brackets (NSNs 4210-01-509-8185 (Part number 130-010) 4210-01-261-5118 (Part Number FH-900-26) or 4210-01-519-0943 (Part Numbers 1299-4292 and 130-042)) shall be verified for compliance with paragraph 3.5.3 following the procedures in the sub-paragraphs of 4.5.4.2.

4.5.4.4 Cylinder bracket retention - ballistics. Fully-charged HHFE systems shall be verified for compliance with paragraph 3.5.4. IAW Federal Specification BB-E-2879A, each fully-charged HHFE shall be mounted in its bracket which shall be securely mounted using its mounting hardware at a distance of 6 ± 1.5 meters (20 ± 5 feet) from the gun muzzle. Each HHFE's cylinder shall be shot with a 0.30-caliber armor piercing cartridge, M2, at a projectile muzzle velocity of 845 ± 20 m/s (2765 ±75 ft/s). Each HHFE shall be impacted with fully tumbled projectiles. A separate HHFE shall be used for each ballistic firing test. HHFEs shall be shot with the longitudinal axis of the cylinder normal to the line of fire. HHFEs shall be shot with the longitudinal axis of the cylinder 45° from normal toward the gun position and the outlet port (neck) of the cylinder facing away from the weapon. HHFEs shall be shot with the longitudinal axis of the cylinder parallel to the line of fire with outlet port (neck) facing away from the gun. Lastly, HHFEs shall be shot with the longitudinal axis of the cylinder normal to the line of fire with the fully tumbled projectile penetrating through the back of the bracket. In any tested orientation, if the cylinder or any fragments from the cylinder larger than 2 inches in diameter do not remain within the envelope of the bracket after impact with the projectile, the test shall be designated a failure.

4.5.4.5 Cylinder fragmentation resistance (non-shatterability).

4.5.4.5.1 Steel (Type I) cylinder. Fully-charged Type I HHFEs shall be verified for compliance with paragraph 3.5.5.1. During this verification each HHFE may be supported but shall not be constrained. While each HHFE is at gun room ambient temperature, its cylinder shall be shot with a tumbling .50-caliber W-2 armor-piercing projectile at a minimum velocity of 2600 feet per second. The HHFE shall be shot from a maximum range of 50 yards. A separate HHFE shall be used for each ballistic firing test. HHFEs shall be shot with the longitudinal axis of the cylinder normal to the line of fire. HHFEs shall be shot with the longitudinal axis of the cylinder 45 degrees from normal to the line of fire toward the gun position. HHFEs shall be shot with the longitudinal axis of the cylinder parallel to the line with the inlet port facing away from the gun position. In any tested orientation, if the cylinder does not remain in one piece or if its resulting dimensional size exceeds 125 percent of its original size, the test shall be designated a failure.

4.5.4.5.2 Aluminum alloy (Type II) cylinder. Type II HHFEs shall be verified for compliance with paragraph 3.5.5.2. Each HHFE's cylinder shall be visibly inspected to verify that the cylinder is marked "NONSHAT" IAW paragraph 3.6.2.4 of Federal Specification RR-C-901 which shall indicate that it is IAW paragraph 3.2.2 of Federal Specification RR-C-901 for non-shatterability. A cylinder not marked accordingly shall constitute a failure of this verification.

4.5.4.6 Safety pull pin/Locking device. HHFEs shall be verified for compliance with paragraph 3.5.6. The discharge handle of each HHFE shall be visually inspected to verify that it includes a safety device, such as a pull-pin, to prevent accidental compression of the handle

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resulting in an unintentional discharge of the HHFE. The lack of a safety device shall constitute a failure of this verification. The safety device shall then be disengaged from the discharge handle by hand and released from grasp. If the safety device does not remain attached to the HHFE or one of its components, via a tether or other similar means, the test shall be designated a failure of this verification.

4.5.5 Performance.

4.5.5.1 Discharge angle and range. Each HHFE tested shall be weighed before and after each test and the results recorded. Fully-charged HHFEs shall be discharged at 70 ± 2°F with the cylinders affixed (no agitation) and oriented at each of the three different angles: -45°, 0° (vertical), and +45°. In every case, the HHFE's swing arm and discharge horn shall be affixed at a height of 4 feet and directed to discharge horizontally. One foot increments shall be marked along the wall in the direction of the HHFE's swing arm and discharge horn to measure the range when viewed from the side perpendicular to the discharge. Each HHFE shall then be discharged until only the pressurization gas escapes the discharge horn. During the discharge, the range distance of the agent shall be observed and recorded. Also, the percent of agent discharged by each HHFE shall be determined by applying the sample HHFE weights recorded before and after discharge to the following formula (Percent Discharged = 100 x [HHFE Wt (grams) As Received - HHFE Wt (grams) After Discharge]/[HHFE Wt (grams) As Received - HHFE Wt (grams) Empty]). Discharge ranges or discharge percentages less than those specified in 3.6.1 shall constitute a failure of this test.

4.5.5.2 Discharge time and amount. Each HHFE shall be weighed before and after each test and the results recorded. Each HHFE evaluated shall initially be conditioned to 21 ± 1°C (70 ± 2°F) for 24 hours. The HHFE shall then be held stationary in a fixture (no agitation) and discharged until pressurization gas begins to escape the discharge horn. Each HHFE's discharge time shall be timed and recorded. The percent of agent discharged by each HHFE shall be determined by applying the sample HHFE's weight recorded before and after discharge to the formula in paragraph 4.5.5.1 above. Any discharge amounting to less than the weight percentage of the agent specified in 3.6.2 constitutes a failed test.

4.5.5.3 Class B fire extinguishment. Successful completion of the flammable liquid pan fire tests described in paragraphs a. and b. below shall be demonstrated. In each case, the pan shall be made of steel, square in shape, and 200 to 305 mm (8 to 12 inches) in depth. The fuel layer in the pans shall be 51 mm (2 inches) deep with its surface located 152 ± 6 mm (6 ± 0.25 inches) below the top edge of the pan. The fuel shall be ignited and allowed to burn for 60 ± 5 seconds prior to each extinguishment attempt. Complete guidelines to be followed for the pan fire tests and pan construction are found in Underwriters Laboratories Standard for Rating and Fire Testing of Fire Extinguishers, UL Standard 711. The 7.5 square foot pan dimensions for construction is not provided in UL Standard 711 and therefore provided in Table III.

a. JP-8 fueled pan fire. A professional firefighter, using a fully-charged HHFE for each of two practice attempts and the following six consecutive attempts, shall successfully perform at least two consecutive, non-intrusive extinguishments of pan fires having a

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fuel surface area of 1.16 square meters (12.5 square feet). Each pan fire shall be fueled with JP-8 aviation turbine fuel purchased from Defense Logistics Agency (DLA) which conforms to MIL-DTL-83133 specifications. Just prior to ignition for each attempt, 50 mL of n-heptane shall be poured in one corner of the pan to assist ignition.

b. Heptane-fueled pan fire. A professional firefighter, using a fully-charged HHFE for each of two practice attempts and the following six consecutive attempts, shall successfully perform at least two consecutive, non-intrusive extinguishments of pan fires having a fuel surface area of 0.70 square meters (7.5 square feet). Each pan fire shall be fueled with commercial-grade heptane having the characteristics described in paragraph 8.2.4 of UL Standard 711.

TABLE III. Fire Pan Sizes

Pan Size (Inside Dimensions)		**Metal Thickness**		**Reinforcing Angle**		**Amount of Commercial Grade heptane or JP-8 used (Approximate)**	
Square Meters	Square Feet	Millimeters	Inches	Millimeters	Inches	Liters	Gallons
0.7	7.5	6	0.25	38 by 38 by 4.8	1.5 x 1.5 x 0.1875	36.7	9.7
1.16	12.5	6	0.25	38 by 38 by 4.8	1.5 x 1.5 x 0.1875	57	15.5

4.5.5.4 Discharge pattern. HHFEs shall be verified for compliance with paragraph 3.6.4. Each HHFE tested shall be fully charged and statically mounted to a fixture having its horn's discharge direction a horizontal distance of 0.61 meters (2 feet) from a backdrop. The backdrop shall be 3.66 meters (12 feet) long and 1.22 meters (4 feet) high having horizontal and vertical lines spaced one foot apart to form a grid. Each grid line on the backdrop shall have additional visible markings at one inch increments. The HHFE's agent shall be discharged parallel to the backdrop until the pressurization gas is observed. Video and photographs shall be taken of the HHFE and the backdrop to determine the discharge diameter.

4.5.6 Operation. The HHFE system shall be verified for operation and use as stated in the below subsections.

4.5.6.1 Human interface. Fully-charged HHFEs clamped in their brackets shall be used to verify compliance to paragraph 3.7.1. One female meeting the physical characteristics of a 5th percentile female aircrew member and one male meeting those of a 95th percentile male aircrew member shall demonstrate this verification while attired in the most cumbersome Air Warrior configuration. Each gender shall successfully perform this verification four times; two demonstrations while wearing properly fitted Cold Weather Flyers' Gloves and two demonstrations while wearing properly fitted Arctic Mittens. Acting alone, each step shall be

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performed as quickly as the aircrew member can. First, each crew member shall disengage the bracket latch that secures the HHFE and remove the HHFE from within the envelope of the bracket. Secondly, the operator shall remove the safety pull pin from the discharge handle. Lastly, the operator shall discharge the HHFE by fully compressing the handle with one hand and sustaining the compression for at least 9 consecutive seconds. Respectively, the following results shall constitute a failed verification: (1) if the time from each crewmember's first contact with the bracket latch until the HHFE is out of the bracket's envelope exceeds 5 seconds; (2) if the time from first contact with the safety pull pin until it is free from the handle exceeds 4 seconds; and (3) if any crewmember cannot sustain full compression of the handle with one hand for 9 consecutive seconds.

4.5.6.2 Bracket unlatching effort. HHFE brackets shall be used to verify the requirement of paragraph 3.7.2. A fully-charged HHFE shall be clamped into a bracket. Using a spring scale, or a similar device, a gradually increasing force shall be applied to the tip of the latch of each bracket strap at a 90-degree angle to the HHFE's longitudinal axis. The force shall be applied at a rate of 270 N (60 lb-f) per minute until the strap releases the HHFE. Only a maximum applied force of 65 N (15 lb-f) or less recorded at the time of the latching strap's release shall constitute a passed test. If the test load reaches 90 N (20 lb-f) without releasing the strap from the HHFE, the test shall be stopped and a failed test indicated.

4.5.6.3 Discharge effort. Fully-charged HHFEs shall be used to verify the requirement of paragraph 3.7.3. The discharge handle of each fully-charged HHFE shall be compressed one time until pressurization gas begins to escape the discharge horn. The forces required to maintain full compression of the handle shall be monitored during the discharge event. If the force required to maintain a momentary (less than 3 second) full compression of the handle exceeds 165 N (37 lb-f) at any time during the discharge event, a failed test shall be indicated. Also, if the required force exceeds 100 N (22 lb-f) for a sustained duration greater than 3 seconds, a failed test shall be indicated.

4.5.6.4 Safety pull pin & tamper indicator. HHFEs and tamper indicators/seals shall be used to verify the requirement of paragraph 3.7.4 according to the procedures in the following sub-paragraphs:

4.5.6.4.1 Tamper indicator. An HHFE and indicators/seals shall undergo this destructive test. Using a secured HHFE, each tamper indicator/seal shall be subjected to a tensile load up to 90 N (20 lb-f) at a rate not to exceed 270 N (60 lb-f) per minute. If the applied load reaches 90 N (20 lb-f) without seal breakage, the test shall be stopped and a failed test indicated. Otherwise, the force applied at breakage shall be recorded, but only a breakage at or below 65 N (15 lb-f) shall be recorded as a passed test.

4.5.6.4.2 Discharge handle loading. HHFEs shall be used to verify the portion of paragraph 3.7.4 to which this verification applies. The safety pull pin shall be fully inserted into the discharge handle of each HHFE which shall be oriented and secured vertically in a fixture. A downward force of 225 N (50 lb-f) shall be exerted for 30 seconds at a distance of 13 mm (0.5 inch) from the end of the discharge handle's top lever. The force shall be applied in a manner which creates a shear load on the inserted pull pin, a manner in which the HHFE would discharge if the pin were not inserted. At this point, if the HHFE discharges or if the pull pin or

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the discharge handle fractures or permanently deforms, a failed verification shall be indicated and the verification process halted. After successful completion of the previous test, a force shall be applied in an effort to pull or dislodge the pull pin from the handle in the intended manner. The pull force exerted shall be applied at a rate no greater than 270 N (60 lb-f) per minute. If the pull pin is not extracted from the handle at or below 135 N (30 lb-f), a failed verification shall be indicated.

4.5.6.4.3 Pull pin extraction. HHFE and safety pull pins shall undergo this verification. For this test, the HHFE shall be oriented and secured vertically in a fixture. Initially, each safety pull pin tested shall be inserted into its intended location in the HHFE's discharge handle. The pull pin shall then be pulled with 135 N (30 lb-f) for no less than 30 seconds at an angle of 45 degrees in an upward direction from the horizontal, and then repeated at a horizontally rotational angle of 45 degrees. To constitute successful verification, the pin shall either: (1) be extracted at the 45 degree angle; or (2) not be damaged or the mechanism jammed to the extent that prevents intended extraction of the pull pin directly outward along its axis when a 135 N (30 lb-f) force is applied for no more than 30 seconds. Note: In all cases, the 135 N force shall be achieved at a rate of no more than 270 N (60 lb-f) per minute.

4.5.7 Environmental conditions. The HHFE system shall be tested as stated in the below subsections.

4.5.7.1 Temperature extremes. The HHFE's ability to withstand the temperature extreme requirements of 3.8.1 shall be verified for compliance with the following subparagraphs.

4.5.7.1.1 Non-operational high-temperature cycles. The HHFE shall be stored in a chamber that shall be conditioned to the representative daily temperature schedule indicated by Table IV below. The HHFE shall be stored continuously for seven of the indicated 24-hour cycles with its temperature being monitored using a calibrated thermocouple attached to the cylinder surface. For the first cycle, the HHFE shall be conditioned to the chamber's initial temperature for a minimum of twenty-four hours. For subsequent hours, it is only necessary to adjust the chamber's temperature setting. A visual or audible observation of HHFE leakage during any cycle shall constitute a failure of this test for the no-leak requirement. A minimum of eighteen HHFE samples shall be tested IAW the above requirements of this paragraph. Upon completion of the seventh cycle, Samples 1 through 8 shall directly be followed by conditioning and testing IAW paragraph 4.5.7.1.2 for high temperature operation. For Samples 9 through 18, the seventh cycle shall instead be followed by the HHFE being conditioned and tested IAW paragraph 4.5.5.2 for discharge time and amount.

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TABLE IV. High Temperature Extreme Cycling

Time of Day	Temperature		Time of Day	Temperature		Time of Day	Temperature	
	(°C)	(°F)		(°C)	(°F)		(°C)	(°F)
0100	36	96	0900	48	118	1700	80	175
0200	34	94	1000	58	136	1800	74	165
0300	34	94	1100	64	148	1900	63	146
0400	33	91	1200	74	165	2000	54	128
0500	33	91	1300	82	180	2100	44	111
0600	33	91	1400	84	183	2200	41	106
0700	37	99	1500	85	185	2300	38	101
0800	43	109	1600	84	183	2400	36	96

4.5.7.1.2 High temperature operation. The HHFE shall be conditioned and stabilized for twenty-four hours at 71°C (160°F). The temperature shall be verified using a calibrated thermocouple to monitor the cylinder surface. Upon stabilizing the HHFE's temperature at 71°C (160°F), the pressure indicator gauge of each HHFE undergoing this procedure shall be visually inspected and photographed to verify that the dial indicator reads between the midpoint and far right of the green zone on the gauge. Gauge accuracy shall be deemed unacceptable if the dial indicates outside of the aforementioned zone. A minimum of eight HHFE samples, numbered 1 through 8, shall be used for testing IAW this paragraph and directly be tested for JP-8 pan fire performance IAW paragraph 4.5.5.3.a, except the HHFE's temperature shall be maintained at 71°C (160°F) until the time of pan fire ignition.

4.5.7.1.3 Non-operational low temperature. The HHFE shall be conditioned to -62°C (-80°F) for a minimum of 72 consecutive hours and verified using a calibrated thermocouple attached to the cylinder surface. A visual or audible observation of HHFE leakage during the 72-hour period shall constitute a failure of this test for the no-leak requirement. The HHFE shall then immediately be verified to maintain its low temperature operational capability IAW 4.5.7.1.4 below. The HHFE shall undergo the requirements of this paragraph, followed by the quantity and performance requirements stated in paragraph 4.5.7.1.4 below.

4.5.7.1.4 Low temperature operation. The HHFE shall be conditioned and stabilized for one hour at -54°C (-65°F). The temperature shall be verified using a calibrated thermocouple to monitor the cylinder surface. Upon stabilizing the HHFE's temperature at -54°C (-65°F), the pressure indicator gauge of each HHFE undergoing this procedure shall be visually inspected and photographed to verify that the dial indicator reads between the far left and midpoint of the green zone. Gauge accuracy shall be defined as a failure if the dial indicates outside of the aforementioned zone. Lastly, the HHFE shall be tested for extinguishment performance IAW paragraph 4.5.5.3 but with the following exceptions: the HHFE's temperature shall be maintained at -54°C (-65°F) until pan fire ignition, and the pan sizes used for the JP-8 and heptane pan fire tests shall be 5 ft^2 and 3 ft^2, respectively. A minimum of sixteen HHFE samples shall undergo the requirements of this paragraph. Samples 1 through 8 shall be performance

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tested against JP-8 fires IAW 4.5.5.3.a, and Samples 9 through 16 tested for performance against n-heptane fires IAW 4.5.5.3.b, which allows for up to two practice runs for each fuel type.

4.5.7.1.5 Temperature shock. The HHFE system shall be verified for compliance with paragraph 3.8.1.5 according to the following procedure:

Step 1. Install a thermocouple on the HHFE cylinder surface in order to monitor thermal response.

Step 2. Mount the HHFE in a temperature controlled test chamber in its deployed configuration and condition the chamber and HHFE to controlled standard ambient conditions.

Step 3. Follow the schedule indicated in FIGURE 5 below, transferring the HHFE between two test chambers, one being used for the low temperature extreme (T1) of -62°C (-79.6°F) and the other for the high temperature extreme (T2) of +85°C (+185°F). During each cycle, be sure to allow the HHFE to stabilize at each temperature extreme prior to its immediate (no more than one minute) transfer to the other extreme environment. Also, when ramping the HHFE and respective chamber to and from controlled ambient conditions, adjust the chamber's temperature at a rate less than 3°C (5°F) per minute.

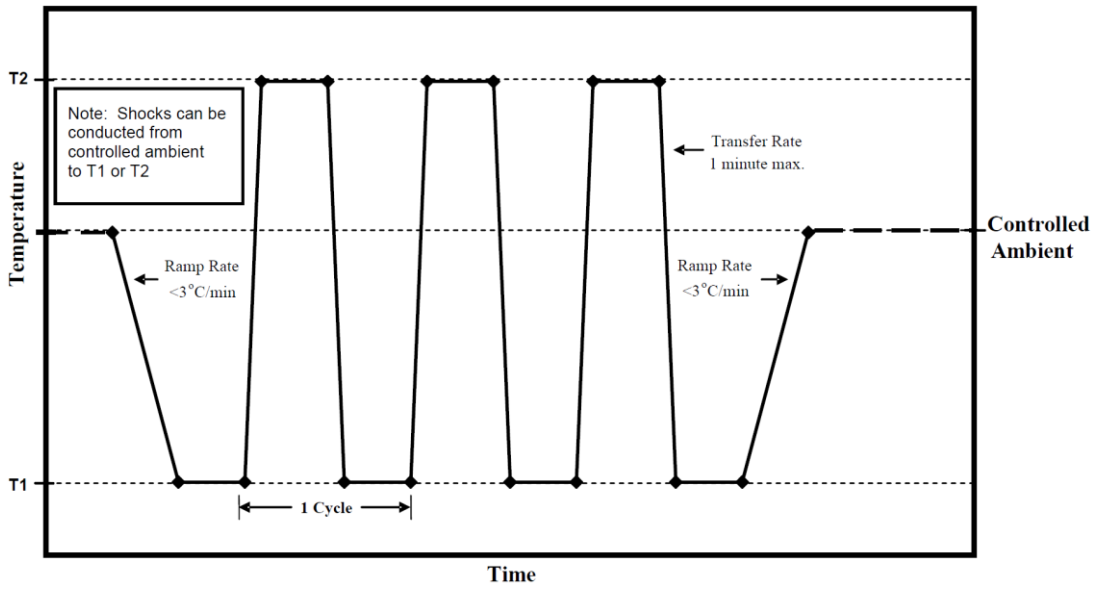


FIGURE 5. Temperature Shock Cycle.

Step 4. Verify the HHFE's performance capability with respect to discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.2 Resistance to high altitude (low pressure) environment. HHFEs shall be verified for compliance with paragraph 3.8.2. HHFEs shall be tested IAW MIL-STD-810, Method 500.5, Procedure I-Storage/Air Transport, using a simulated non-operational test altitude of 15,240 meters (50,000 feet) for a minimum of one hour. Afterward, the HHFEs' performance shall be

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verified for discharge time and amount IAW paragraph 4.5.5.2 at standard ambient conditions. HHFEs shall be tested IAW MIL-STD-810, Method 500.5, Procedure II – Operation/Air Carriage. A simulated operational test altitude of 6,096 meters (20,000 feet) at the standard ambient temperature and humidity shall be used for the test. After holding each HHFE in the test chamber at the simulated test altitude for a minimum of one hour, the HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2. Although the HHFE hardware shall remain exposed to the simulated test altitude during the performance check, the agent may be discharged into standard ambient conditions.

4.5.7.3 Humidity. HHFEs clamped within their brackets shall be verified for compliance with paragraph 3.8.3 IAW this paragraph. IAW MIL-STD-810, Method 507.5, Procedure II (Aggravated), each HHFE system shall be placed in a chamber and conditioned to 23 ± 2°C (73 ± 4°F) at 50 ± 5 percent relative humidity and maintained for no less than 24 hours. The chamber shall be adjusted to 30°C (86°F) and its relative humidity to 95 ± 4 percent. The chamber's temperature shall then be controlled according to FIGURE 6 below for ten of the cycles shown. At the completion of the 10th cycle, adjust the temperature and humidity to 23 ± 2°C (73 ± 4°F) and 50 ± 5 percent relative humidity, and maintain until the test item has reached temperature stabilization (generally not more than 24 hours). After exposure to the aforementioned test methods, each HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

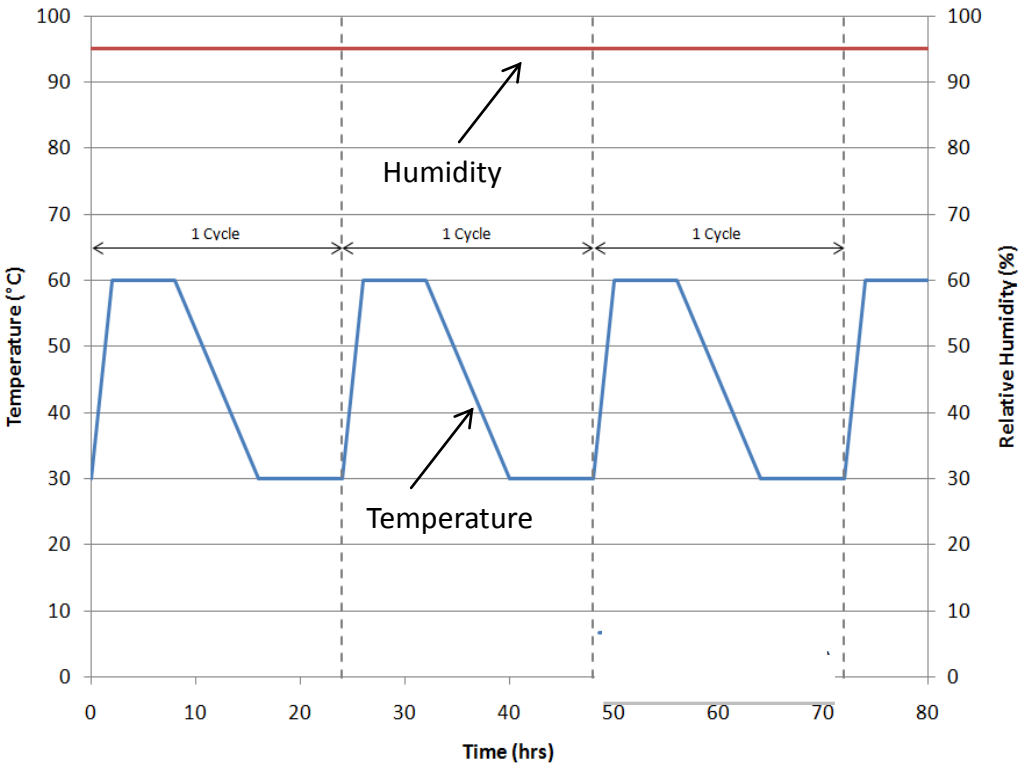


FIGURE 6. Aggravated temperature-humidity cycle.

Notes:
1. Maintain the relative humidity at 95 ± 4 percent at all times except that during the descending temperature periods the relative humidity may drop as low as 85 percent.
2. A cycle is 24 hours.

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4.5.7.4 Fungus. HHFEs clamped within their brackets shall be verified for compliance with paragraph 3.8.4. All verification methods, such as the cleaning and handling of the test sample, preparation of the mineral salt solution and the mixed fungus spore suspension, the preparation of the control items, and the test and evaluation methods shall be IAW Test Method 508.6 of MIL-STD-810. As for the HHFE and bracket's incubation period in Test Method 508.6, each tested assembly shall be incubated in the test chamber for fungus growth for no less than 28 days at 30 ± 1°C (86 ± 2°F) and a relative humidity of at least 90 percent but less than 100 percent. Following the incubation period each HHFE shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.5 Salt fog. HHFEs clamped within their brackets shall be verified for compliance with paragraph 3.8.5. Methods used for verification, such as test sample handling, salt solution, test and evaluation methods, shall be IAW Test Method 509.5 of MIL-STD-810. After exposure to the aforementioned test methods, each HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.6 Sand and dust environment. The HHFE shall be verified for compliance with paragraph 3.8.6. Methods used for verification, such as the test chambers' dust and sand concentrations, test and evaluation methods, shall be IAW Test Method 510.5 of MIL-STD-810. HHFE assemblies shall be tested to Method 510.5's Procedure I-Blowing Dust followed by each HHFE being verified for discharge time and amount IAW paragraph 4.5.5.2. A separate quantity of HHFE assemblies shall be tested to Procedure II-Blowing Sand followed by each HHFE being verified for discharge time and amount IAW paragraph 4.5.5.2. The assemblies shall be conditioned to the high operating temperature stated in paragraph 3.8.1 during its exposure to the blowing sand in the test chamber.

4.5.7.7 Acidic atmosphere. HHFE systems shall be verified for compliance with paragraph 3.8.7. Methods used for verification, such as the required acidic concentration for the test chamber atmosphere, testing and evaluation methods, shall be IAW MIL-STD-810, Method 518.1. The HHFE system shall undergo 4 periods in an acidic atmosphere test chamber with each period being followed immediately by 7 days of storage at standard ambient conditions. For each period of acidic atmosphere exposure, the HHFE shall be clamped within its bracket and conditioned to 35°C (95°F) within the test chamber. Then, the chamber shall be injected for 2 hours with an atomized test solution consisting of 11.9 mg sulfuric acid and 8.8 mg nitric acid for every 4 liters of solution in distilled water. For each of the four periods, the acid solution shall be preheated according to the schedule indicated in Table V prior to be injected into the test chamber. After exposure to the aforementioned test methods and the last storage period, each HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

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TABLE V. Temperature and Pressure Requirements for operation at 35°C.

Air Pressure (kPa)	83	96	110	124
Preheat temperature (°C) (before atomizing)	46	47	48	49

4.5.7.8 Functional shock. HHFEs shall be verified for compliance with paragraph 3.8.8 IAW this paragraph. Each HHFE shall be subjected to the test shock input indicated in Table VI and FIGURE 7 below IAW the methods and criteria of MIL-STD-810, Method 516.6, Procedure I. Each HHFE shall be verified for no discharge of its contents after being subjected to each functional shock by visual inspection around the head mechanism, inside and around the horn and nozzle areas. The observed presence of extinguishing agent residue, such as SBC_S powder, shall constitute a failure of this verification. Afterward, each HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

TABLE VI. Test shock response spectra (taken from Table 516.6-I in MIL-STD-810 Method 516.6 Paragraph 2.3.2.b.)

Test Category	Peak Acceleration (g's)	T_e (ms) /1	Cross-over Frequency (Hz)
Functional Test for Flight Equipment	20	15-23	45

Note 1: To customize the bandwidth of the Shock Response Spectrum (SRS) and T_e, refer to guidance in paragraph 2.3.3 c and d of MIL-STD-810 Method 516.6.

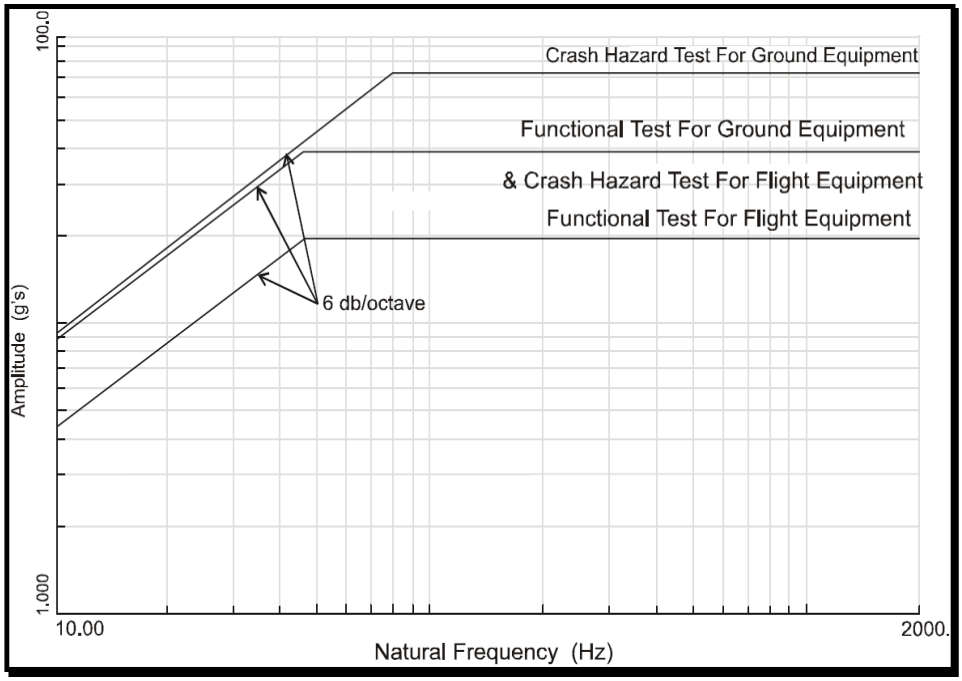


FIGURE 7. Test SRS (taken from Figure 516.6-8 in MIL-STD-810 Method 516.6 Paragraph 2.3.2.b.)

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4.5.7.9 Bench handling shock. HHFEs shall be verified for compliance with the requirements of paragraph 3.8.9 IAW this paragraph. IAW MIL-STD-810, Method 516.6, Procedure VI, each HHFE shall undergo four bench shocks on a steel work bench surface. To administer each shock, use the HHFE's bottom edge as a pivot, then tilt the top shoulder of the cylinder to a vertical height of 100 mm (4 inches) above the surface and release it. Each shock shall be administered with the HHFE's horn positioned against the side of the HHFE's cylinder. The first shock shall be administered such that the table's surface shall first contact the HHFE's horn squarely. The other three shocks shall be administered relative to the first at 90°, 180°, and 270°, measured with the HHFE being rotated about its longitudinal axis. Each HHFE shall be verified for no discharge of its contents after being subjected to the fourth shock by a visual inspection around the head mechanism and inside the horn. The observed presence of extinguishing agent residue, such as SBC_S powder, shall constitute a failure of this verification. Afterward, each HHFE's performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.10 Crash hazard shock. The HHFE system shall be verified for compliance with paragraph 3.8.10 IAW MIL-STD-810, Method 516.6, Procedure V. Two 40g shocks shall be administered in each direction along three orthogonal axes of the HHFE for a maximum of 12 shocks. Each of the 12 shocks shall be performed while the HHFE is mounted in its bracket. The following occurrences with respect to the bracket shall constitute a failure of this verification: a bracket fails to retain the HHFE within its clamp, and/or the bracket containing the HHFE is released from the bracket's mount. Each HHFE shall be verified for no discharge of its contents after being subjected to each shock by a visual inspection around the head mechanism area and inside the horn. The observed presence of extinguishing agent residue, such as SBC_S powder, shall constitute a failure of this verification. After each tested HHFE's 12th and final shock, its performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.11 Vibration. The HHFE shall be verified for compliance with paragraph 3.8.11 IAW this paragraph. A modal survey of the HHFE system and the expected platforms shall be performed. Vibration testing IAW MIL-STD-810, Method 514.6 shall be conducted on each HHFE system using the appropriate vibration spectrum based on the worst-case platform requirements as derived from the modal survey. Vibration events shall be applied to the HHFE/bracket assembly with the bracket mounted to a test fixture. Each HHFE shall be verified for no discharge of its contents after being subjected to the vibrations by a visual inspection inside the horn. The observed presence of extinguishing agent residue, such as SBC_S powder, shall constitute a failure of this verification. An HHFE becoming unclamped from its bracket during the application of test vibrations shall also constitute a failure of this verification. After each HHFE's final vibration exposure, its performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.12 Drop test. HHFEs, without the bracket, shall be verified for compliance with paragraph 3.8.12 IAW this paragraph. IAW MIL-STD-810, Method 516.6, Procedure IV, each HHFE shall be dropped six times onto 2-inch plywood backed by concrete from a height of four feet. The six drops per HHFE shall consist of two drops with the longitudinal axis parallel to the surface, two drops with the longitudinal axis perpendicular to the surface with the extinguisher bottom pointing down, and two drops with the longitudinal axis perpendicular to the surface with the extinguisher bottom pointing up. The impact point or surface for each drop shall be

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documented. Any visually observed damage to the HHFE and/or its components shall be documented, including the specific area or component damaged and whether the HHFE can be subsequently tested for discharge time and amount as stated below. Each HHFE shall be verified for no discharge or leak of its contents after each drop by a visual inspection inside the horn and around the components that are under pressure. The observed presence of extinguishing agent residue, such as SBC_S powder, shall constitute a failure of this verification. Any observed leaking or leakage of the HHFE's contents or any observed damage to pressurized components shall also constitute a failure of this verification. After each HHFE's sixth and final drop is administered, its performance shall be verified for discharge time and amount IAW paragraph 4.5.5.2.

4.5.7.13 Aging resistance of rubber materials (horn). HHFE horns shall be verified for compliance with paragraph 3.8.13 IAW this paragraph. The horn shall be placed within a pressure bomb. The bomb shall then be pressurized with oxygen at 290 to 310 psig and placed in an oven. The oven shall then be heated to 71°C (160°F) and maintained for 24 hours. The bomb and horn shall be allowed to cool, and then the pressure released slowly. Any observed fractures or cracks to the rubberized horn materials resulting from this test shall constitute a failure of this verification.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Definitions.

6.1.1 "Withstand" and/or "survive" - For the purposes of this specification's requirements, the terms "withstand" and "survive" are defined as the following:

- The HHFE does not leak its contents to the surrounding environment
- The HHFE's cylinder or any other component does not rupture or permanently deform from the pre-test shape/size
- The HHFE, its components and/or its contents do not degrade such that they cannot meet any of the requirements stated in this document

6.1.2 Inspection – Inspection is the process of measuring, examining, testing, or otherwise comparing the unit of product (see definition 6.1.3 below) with the requirements.

6.1.3 Unit of Product – The unit of product is the unit inspected in order to determine its classification as conforming or nonconforming or to count the number of nonconformities. It may be a single article, a pair, a set, a length, an area, an operation, a volume, a component of an end product, or the end product itself. The unit of product may or may not be the same as the unit of purchase, supply, production, or shipment.

6.1.4 Agent. Agents are classified as Class I or Class II and are free-flowing slurries of either of two specialized sodium bicarbonate powders suspended in HFC-227ea that will be used as Agent in fire extinguishing hardware

6.1.5 Slurry. Either SBC-1 or SBC-2 powder in liquid HFC-227ea as a free-flowing suspension.

6.1.6 Agent class. The Agent is either a Class I or Class II Agent with Class I Agent being a slurry of HFC-227ea and SBC-1 and Class II Agent being a slurry of HFC-227ea and SBC-2.

6.1.7 Acronyms.

ANSI	=	American National Standards Institute
ASTM	=	American Society for Testing and Materials
°C	=	degree(s) Celsius
CARC	=	Chemical Agent Resistant Coating
CC	=	cubic centimeter(s)
CF_3CHFCH_3	=	Heptafluoropropane
CFR	=	Code of Federal Regulations
CID	=	Commercial Item Description(s)
cm	=	centimeter(s)
CO_2	=	Carbon Dioxide
DLA	=	Defense Logistics Agency

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DOT	=	Department of Transportation
°F	=	degree(s) Fahrenheit
ft	=	foot (or feet)
ft^3	=	cubic foot (or foot cubed) (plural-see ft)
g	=	gravity
Halon 1301	=	Bromotrifluoromethane **(**CF_3BR)
HFC-227ea	=	1,1,1,2,3,3,3-Heptafluoropropane agent
HFC-227ea/SBC_S	=	HFC-227ea Agent Enhanced with SBC_S
HHFE	=	Handheld Fire Extinguisher
Hz	=	hertz
IAW	=	in accordance with
lb-f	=	pounds-force
k	=	kilo
kPa	=	kilopascal(s)
m	=	meter(s)
m^3	=	cubic meter(s) (or meters cubed)
mm	=	millimeter(s)
mph	=	miles per hour
ms	=	millisecond(s)
MSDS	=	Material Safety Data Sheet
N	=	Newton(s)
NFPA	=	National Fire Protection Agency
NONSHAT	=	shatterproof / non-shatter
NSN	=	National Stock Number
ODS	=	ozone-depleting substance
Pa	=	Pascal(s)
PSI (or psi)	=	pounds per square inch
psig	=	psi - gauge
ref	=	reference
RH	=	relative humidity
s	=	second(s)
SBC_S	=	Special Sodium Bicarbonate
SRS	=	Shock Response Spectrum
TDP	=	Technical Data Package
UL	=	Underwriters' Laboratory

6.2 Pressurization gas verification alternative. In lieu of analysis, a certificate of analysis is recommended from the manufacturer that Commercial Item Description (CID) A-A-59503 Type I, Grade A, class 1 nitrogen is used to pressurize the HHFE.

6.3 Hexavalent chrome verification. A standard for verification of the lack of Hexavalent chromates' use in pretreatments does not exist. The manufacturer is to provide a written certification that Hexavalent chromates are not used in metal substrate pretreatments of HHFE system components to include Material Safety Data Sheets (MSDSs) for the pretreatments used.

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6.4 Chemical Agent Resistant Coatings. In lieu of analysis, written verification is recommended from the manufacturer that coatings conforming to the requirements of MIL-DTL-53072 are used.

6.5 Acquisition requirement. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Packaging requirements (see 5.1)
(c) When first article inspection is required (see 3.1)
(d) Degree of preservation and degree of packing required (see 5.1)

6.6 Additional Guidance. The following Department of Defense Standards and non-government publications listed below provide additional guidance as needed.

MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-1916	Preferred Methods for Acceptance of Product
MIL-STD-2073	Standard Practice for Military Packaging
ASTM E171	Standard Practice for Conditioning and Testing Flexible Barrier Packaging
NFPA 10	Standard for Portable Fire Extinguishers

6.7 Requirements & verifications cross reference. Table VII provides a cross reference between the Requirements in Section 3 and the respective verification methods in Section 4 of this specification.

TABLE VII. Requirement-to-Verification Cross Reference Matrix

Requirement	Verification	Title
3.2	**4.5.1**	**Materials and Components**
3.2.2	4.5.1.1	Fire extinguishing agent (HFC 227ea/SBC_S)
3.2.3	4.5.1.2	Agent exposure
3.2.4	4.5.1.3	Pressurization gas
3.2.5	4.5.1.4	Coatings and surface finishes
3.2.6	4.5.1.5	Discharge handle
3.2.7	4.5.1.6	Pull pin/Safety locking device
3.2.8	4.5.1.7	Horn
3.3	**4.5.2**	**Construction**
3.3.1	4.5.2.1	Cylinder
3.3.1.1	4.5.2.1.1	Steel (Type I) cylinder
3.3.1.2	4.5.2.1.2	Aluminum alloy (Type II) cylinder
3.4	**4.5.3**	**Design**
3.4.1	4.5.3.1	HHFE system

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3.4.2	4.5.3.2	HHFE
3.4.3	4.5.3.3	Cylinder
3.4.4	4.5.3.4	Bracket
3.4.5	4.5.3.5	HHFE contents and pressurization
3.4.6	4.5.3.6	Discharge indicator gauge
3.4.7	4.5.3.7	Mounting bracket latch
3.4.8	4.5.3.8	HHFE removal
3.4.9	4.5.3.9	Nameplate
3.4.10	4.5.3.10	Weight
3.5	**4.5.4**	**Safe Design**
3.5.1	4.5.4.1	Safe to handle
3.5.2	4.5.4.2	Non-legacy bracket
3.5.3	4.5.4.3	Legacy bracket compatibility
3.5.4	4.5.4.4	Cylinder bracket retention - ballistics
3.5.5	4.5.4.5	Cylinder fragmentation resistance (non-shatterability)
3.5.5.1	4.5.4.5.1	Steel (Type I) cylinder
3.5.5.2	4.5.4.5.2	Aluminum (Type II) cylinder
3.5.6	4.5.4.6	Safety pull pin/Locking device
3.6	**4.5.5**	**Performance**
3.6.1	4.5.5.1	Discharge angle and range
3.6.2	4.5.5.2	Discharge time and amount
3.6.3	4.5.5.3	Class B fire extinguishment
3.6.4	4.5.5.4	Discharge pattern
3.7	**4.5.6**	**Operation**
3.7.1	4.5.6.1	Human interface
3.7.2	4.5.6.2	Bracket unlatching effort
3.7.3	4.5.6.3	Discharge effort
3.7.4	4.5.6.4	Safety pull pin & tamper indicator
3.8	**4.5.7**	**Environmental Conditions**
3.8.1	4.5.7.1	Temperature extremes
3.8.1.1	4.5.7.1.1	Non-operational high temperature
3.8.1.2	4.5.7.1.2	High temperature operation
3.8.1.3	4.5.7.1.3	Non-operational low temperature
3.8.1.4	4.5.7.1.4	Low temperature operation
3.8.1.5	4.5.7.1.5	Temperature shock
3.8.2	4.5.7.2	Resistance to high altitude (low pressure) environment
3.8.3	4.5.7.3	Humidity
3.8.4	4.5.7.4	Fungus
3.8.5	4.5.7.5	Salt fog
3.8.6	4.5.7.6	Sand and dust environment
3.8.7	4.5.7.7	Acidic atmosphere
3.8.8	4.5.7.8	Functional shock
3.8.9	4.5.7.9	Bench handling shock

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3.8.10	4.5.7.10	Crash hazard shock
3.8.11	4.5.7.11	Vibration
3.8.12	4.5.7.12	Drop shock resistance
3.8.13	4.5.7.13	Aging resistance of rubber materials (horn)

6.8 Subject term (key word) listing.

Fire extinguishing agent
Fire suppression agent
HHFE
Handheld Fire Extinguisher
Hydrostatically Tested Cylinder

6.9 This specification establishes the requirements and verification methods for hardware that can utilize one of two HFC 227ea/SBC_S fire extinguishing agents. It does not provide test methods for approving alternate agent raw materials in this hardware. Approval of any replacement materials in this hardware will require a complete revalidation per the Detail Test Plan.

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EXTINGUISHING AGENT SAMPLING TECHNIQUE FOR ANALYSIS IAW MIL-DTL-32412 AND PROCEDURE FOR DETERMINING SBC_S CONCENTRATION IN HFC-227ea

A.1 SCOPE

A.1.1 Scope. These procedures were prepared for this specification so that sample agent slurry (HFC-227ea/SBC_S mixture) and SBC_S (Class I or Class II) could be obtained from fully-charged HHFEs for conformance analysis IAW MIL-DTL-32412. MIL-DTL-32412's analyses verify that the agent slurry and SBC_S meet parameters for moisture content, caking tendency, lumping tendency, flow, suspendability in HFC-227ea, color, particle aspect ratio (length/width), powder surface area, and all particle size requirements. One procedure provides the steps to verify the weight percentage concentration of either Class I or Class II SBC_S in HHFEs.

A.1.2 Applicability. The procedures specified in this appendix are required. The purpose of these procedures are to describe methods for extracting samples (agent slurry and SBC_S) for the determination of the slurry's characteristics as provided in the HHFE. The procedures define the equipment required for executing the steps for carrying out the extraction of slurry samples for analysis and the controls for insuring the quality of the results. Results shall be reported IAW MIL-DTL-32412. One procedure is required to determine the weight percentage concentration of either Class I or Class II SBC_S in HHFEs.

A.2 APPLICABLE DOCUMENTS:

MIL-DTL-32412	HFC-227ea FIRE EXTINGUISHING AGENT ENHANCED WITH SPECIAL SODIUM BICARBONATE POWDER (HFC-227ea/SBC_S)

A.3 REQUIREMENTS

A.3.1 Sample extraction procedures. The following steps are provided to accurately extract slurry samples for analysis. One procedure detailed in the following shall be used to obtain a slurry sample(s) for analysis of suspendability in HFC-227ea, caking tendency, lumping tendency and flow IAW with MIL-DTL-32412. The other procedure detailed in the following steps shall be used to obtain SBC_S samples to determine SBC_S moisture content, color, particle aspect ratio (length/width), powder surface area, and all particle size requirements IAW MIL-DTL-32412.

A.3.1.1 HHFE agent slurry sampling procedure. Sampling of the slurry shall be done only on HHFEs cooled to below ambient room temperatures (not to exceed 75 °F). All received HHFEs for testing will be assumed to be over-pressured up to 250 psig with nitrogen. The procedure shall follow steps 1 through 5 below.

1. Remove the nozzle and horn from the HHFE swing arm.
2. Shake the HHFE for a period of not less than 30 seconds to assure uniform suspension of its contents.

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3. Immediately reduce the HHFE nitrogen overcharge pressure (initially as high as ~250 psi) by inverting the HHFE, directing the swing arm toward the floor and discharging the nitrogen overpressure gas through the swing arm. Verify that the HHFE's pressure is below 65 psi using a pressure gauge and fittings as needed. Do not proceed further if the HHFE's pressure reading is over 65 psi.

4. Evacuate (pull a vacuum) an Andrews Glass pressure tube (3 oz pressure tube, Andrews Glass part no. 110108003, working pressure = 225 psi) and close its valve. Attach the Andrews Glass pressure tube (or equivalent) to the HHFE swing arm and fill the glass pressure tube to no more than 80% of its height with slurry.

5. Disconnect the pressure tube from the HHFE swing arm and proceed to evaluate the slurry contents within the glass pressure tube for suspendability in HFC-227ea, caking tendency, lumping tendency and flow IAW MIL-DTL-32412 and record the results. Samples that do not meet the parameters of MIL-DTL-32412 will be deemed unacceptable.

A.3.1.2 HHFE SBC_S sampling procedure. Sampling of the agent slurry shall be done only on HHFEs cooled to below ambient room temperatures (not to exceed 75 ^{o}F). All received HHFEs for testing will be assumed to be over-pressured up to 250 psig with nitrogen. The procedure shall follow steps 1 through 5 below.

1. Remove the nozzle and horn from the HHFE swing arm.

2. Shake the HHFE for 30 seconds.

3. Immediately reduce the HHFE nitrogen overcharge pressure (initially as high as ~250 psi) by inverting the HHFE, directing the swing arm toward the floor and discharging the nitrogen over-pressure gas through the swing arm. Verify that the HHFE's pressure is below 65 psi using a pressure gauge and fittings as needed. Do not proceed further if the HHFE's pressure reading is over 65 psi.

4. Attach a valve and filter housing (filter installed) to the fully-charged HHFE's swing arm.

5. Shake this assembly such that the HHFE can discharge liquid and gaseous HFC-227ea (containing SBC_S) from the swing arm through the filter housing with the filter housing outlet facing downward.

6. Transfer the dry powder still contained in the filter housing (see dry box procedure) into an open pre-weighed jar and allow it to further "dry" (lose HFC-227ea absorbed onto particle surfaces) while inside the dry-box.

7. Once HFC-227ea loss is complete, tighten the jar lid, and seal and weigh the jar for shipment to the laboratory to perform water analyses, Field Emission Scanning Electron Microscopy (FESEM), and or particle size distribution analysis IAW with MIL-DTL-32412. NOTE: A minimum sample of 10 grams of SBC_S is required for water analyses and a minimum sample of 1 gram of SBC_S is required for all other analyses.

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A.3.1.3 Dry box test method. Removal of SBC_S (Class I or Class II) and HFC-227ea from the agent slurry shall utilize the dry box method or equivalent. The dry box method requires the use of procedures which maintain a relative humidity of less than 1%, special handling and specialized equipment designed to preserve the SBC_S characteristics and sample water content for analysis.

A.3.1.4 Determining SBC_S concentration in HFC-227ea. The procedure shall be performed only on HHFEs cooled to below ambient room temperatures (not to exceed 75 °F). All received HHFEs for testing will be assumed to be over-pressured to 250 psig with nitrogen. To ensure the most accurate measure of percentage of SBC_S content in an HHFE, the entire HHFE's contents shall be weighed and initial and final weight measurements taken at various points as detailed in the following steps 1 through 13 below.

1. Weigh the test HHFE to the nearest gram. Record this weight as "HHFE initial weight".

2. Reduce the HHFE nitrogen overcharge pressure (initially ~250 psi) by inverting the HHFE, directing the swing arm toward the floor and discharging the nitrogen overpressure gas through the swing arm.

3. With the HHFE in an inverted orientation, continue to vent the HHFE. Allow the HFC-227ea vapor to be vented for at least 12 hours. A clamp may be attached to the valve to keep it open during this operation.

4. Dry the HHFE of any surface water caused by condensation of humid room air.

5. Weigh the dried HHFE to the nearest gram and record the data as "HHFE plus SBC_S". At this point only the SBC_S remains inside the HHFE. Record this weight as the "Weight of HHFE plus SBC_S".

6. Subtracting the "HHFE plus SBC_S" weight (step 5) from the "HHFE initial weight" value yields the weight of HFC-227ea initially contained in the HHFE. Record this value as "HFC-227ea weight".

7. Remove the HHFE valve assembly from the HHFE cylinder. Shake the cylinder and tap the exterior to remove as much SBC_S as possible.

8. Using compressed air, blow out as much residual SBC_S as possible from the cylinder and the valve assembly.

9. Pour 500 mL of hot water into the cylinder and rotate it to contact all surfaces to remove any residual SBC_S. Pour out the hot water and add a second 500 mL of hot water to repeat the rinse. Shake out as much residual water as possible and place the cylinder into an oven with a temperature of 130 °C for 4 hours to bake out all remaining water. At the end of the 4-hour drying period, remove the cylinder and allow it to cool to room temperature.

10. Once the dry cylinder has returned to room temperature, reattach the valve assembly and weigh the assembled HHFE to the nearest gram. Record this weight

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as the "Empty HHFE weight".

11. Subtract the "Empty HHFE weight" (determined in step 10) from the weight of "HHFE plus SBC_S" (determined in step 5) to determine the "Weight of SBC_S in slurry" contained in the HHFE. Record this weight as the "Weight of SBC_S in slurry".

12. Divide the "Weight of SBC_S in slurry" (step 11) by the sum of the HFC-227ea weight plus the "Weight of SBC_S in slurry" and multiply by 100. This is the value for the "percent by weight SBC_S" in the agent slurry.

13. If the value for "percent by weight SBC_S" calculated in 11 above is outside the range for either Class I or Class II SBC_S, the test is failed. If the value is outside the range for SBC-1, the test is failed. Record a pass or fail as appropriate.

A.3.2 Special Equipment and Materials. The equipment and materials listed below or its equivalent shall be required to perform sample extraction procedures and for gathering samples for analysis IAW MIL-DTL-32412.

1. Nano particle filter (Donaldson Tetratex® TX6614 (AX03-767), 50-nanometer membrane bonded to polyester substrate) and filter housing with compatible fittings and valves.
2. A dry box for sample handling (Nitrogen or Argon glove box)
3. 3-oz pressure tube, Andrews Glass part no. 110108003, working pressure = 225 psi
4. High purity air, helium and hydrogen, or equivalent
5. Tools to remove the nozzle from the swing arm and make other connections
6. Sealable sample containers (for preventing sample contamination and shipment of SBC_S for analysis).
7. Scale capable of weighing to 5 kg or greater with weight readings to the nearest 0.1 gram
8. Pressure gauge
9. Fittings and connection tubing

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CONCLUDING MATERIAL

Custodian:
Army-AV
Navy-YD

Preparing activity:
Army-AV
(Project 4210-2012-002)

Review Activities:

Army-AT, MR, AR
Navy-AS
DLA-DH, GS3, IS

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